

North American Bat Monitoring Program in SE Alaska

Data Summary and Activity Trend Analyses: 2017 - 2025

Camila Hurtado

Taylor Hart

2026-10-02

Table of contents

1	Executive Summary	3
2	Introduction	4
2.1	Overview of NABat and the NNW Bat Hub	4
2.2	Threats to bat populations in S.E. Alaska	4
2.3	Acoustic Monitoring of Bats	6
2.4	Statistical Models	6
3	Methods	7
3.1	Data Collection	7
3.1.1	Stationary Surveys	7
3.1.2	Transect Surveys	7
3.1.3	Landscape and Weather Covariates	7
3.1.4	Manual Verification	8
3.2	Statistical Analyses	9
3.2.1	Activity Trends	9
3.2.2	Transect Abundance Models	10
3.2.3	Dynamic Occupancy	11
4	Results	12
4.1	Activity and Abundance Trends	14
4.1.1	Covariate Effects	15
4.2	Dynamic Occupancy	16
4.2.1	Covariate Effects	17
5	Discussion	19
5.1	Interpreting Alaska estimates vs. BC's contributions	19
5.2	Trend Results	19
5.3	Looking forward	20
6	Appendix A	22
6.1	Silver-haired Bat (<i>Lasiyonicteris noctivagans</i>)	22
6.2	California Myotis (<i>Myotis californicus</i>)	24
6.3	Long-eared Myotis (<i>Myotis evotis</i>)	26
6.4	Little Brown Myotis (<i>Myotis lucifugus</i>)	28
6.5	Long-legged Myotis (<i>Myotis volans</i>)	30
7	Appendix B	32
	Literature Cited	71

1 Executive Summary

The North American Bat Monitoring (NABat) program is a multi-agency initiative administered by the US Geological Survey (USGS). In 2024, the North by Northwest (NNW) Bat Hub was formed by combining the former Alberta Hub and the BC and Southeast Alaska (BC-SE_AK) Hub. The NNW Bat Hub leads the NABat program within S.E Alaska, with key support from state representatives. In 2024, 9 grid cells were monitored and in 2025 6 grid cells were monitored in Southeast Alaska. We used activity, abundance and dynamic occupancy models to evaluate the trends in bat population in the region using 7 years of acoustic data. Results indicate that bat populations across Southeast Alaska remain stable, with some key species showing small positive trends, including Little Brown Myotis and Silver-haired bat. Given growing threats to bat populations across the region, continued monitoring is essential to detect any significant changes.

2 Introduction

2.1 Overview of NABat and the NNW Bat Hub

North American Bat Monitoring Program (NABat) is a large-scale coordinated effort to monitor bat species to address gaps in knowledge and lack of long-term studies across North America (Loeb et al. 2015). NABat uses a spatially-balanced random sampling design and standardized protocols for conducting short-term acoustic surveys and colony counts (Loeb et al. 2015). The continental program is administered by the US Geological Survey (USGS). The Alaska Department of Fish and Game, with assistance from the US Forest Service, initiated an NABat monitoring program in Southeast Alaska in 2019.

NABat relies on regional hubs to help coordinate and standardize monitoring across smaller areas. Regional hubs help facilitate local implementation of NABat and maximize the benefits of the program within each region. The North of Northwest (NNW) Bat Hub was established in 2024 with collaboration from the governments of Alaska, Alberta, and British Columbia. The hub was formed by combining two previous regional hubs: the Alberta Hub and the BC and Southeast Alaska Hub. The NNW Bat Hub is directed by a steering committee consisting of representatives from each jurisdiction, major stakeholders, and USGS team members and it is coordinated by Biodiversity Pathways. The NNW Hub is responsible for compiling data, identifying calls to species, conducting statistical analyses, and generating annual regional reports. NABat monitoring in Southeast Alaska is implemented by the Alaska Department of Fish and Game and US Forest Service.

2.2 Threats to bat populations in S.E. Alaska

North American bat populations are currently facing multiple anthropogenic threats, including logging and other forms of habitat loss and fragmentation, disease, pesticides, wind energy development, and climate change. The most widespread and deadly threat is white-nose syndrome (WNS), a fungal disease that attacks hibernating bats. Although WNS is not yet present in Alaska, it has spread as far west as the Rocky Mountains and as far north as the Northwest Territories in Canada (Figure 1). Following its discovery in King County in 2016, WNS has spread throughout most of Washington state and into Oregon (Figure 1) and the fungus (though not the disease) has been detected along the Washington border in southern British Columbia (White Nose Syndrome Response Team 2026).

The impacts of white-nose syndrome have been most severe east of the Rocky Mountains, where bats hibernate in large aggregations in caves and mines, resulting in high mortality (Cheng et al. 2021). In the Rocky Mountains and further west, including in Alaska, Little Brown Myotis hibernate singly or in small groups in holes in the soil or between rocks in talus deposits (Neubaum 2018; Blejwas et al. 2021). This very different hibernation behavior is likely responsible for the slower rate of spread of WNS observed in Washington state (Blejwas et al. 2023). Smaller and more dispersed winter colonies may reduce mortality relative to eastern North America, but will also make it difficult to detect the arrival and spread of WNS in Alaska and track its impact on bats in the state (Blejwas et al. 2023). Given the absence of large hibernacula where bats may be counted and swabbed for the presence of WNS, alternative methods for monitoring bats in Southeast Alaska are required to assess changes or impacts from WNS or other threats.

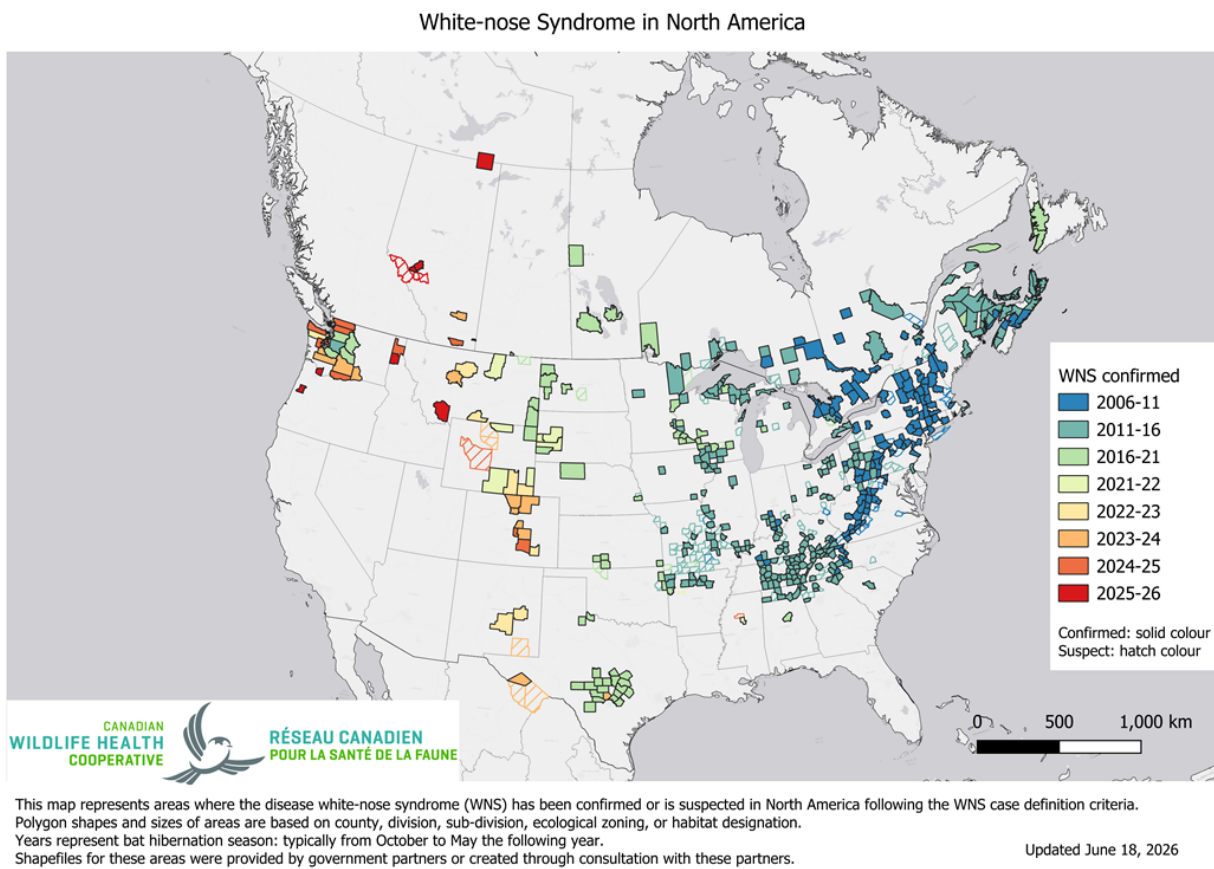


Figure 1: White-nose syndrome (WNS) and *Pseudogymnoascus destructans* (Pd) occurrence in The US and Canada. Source: CWHC-RCSF (2026)

2.3 Acoustic Monitoring of Bats

Acoustic recordings are a cost-effective method for monitoring bat populations across large geographic and temporal scales. However, several limitations affect the conclusions that can be drawn from these data and trend estimates cannot be produced for all species using acoustic methods alone:

1. **Imperfect species identification** — Each bat species can produce a broad range of echolocation calls, corresponding to different behaviours in response to environmental features. Species with similar ecology and physiology often produce overlapping calls, particularly in areas with high clutter (obstacles), such as forested habitats. Although detectors are deployed to maximize recordings of diagnostic (typically Search Phase calls), species identifiability will always vary depending on the local species assemblage.
2. **Variable detectability** — Bat species are not equally detectable acoustically. Larger, wide-ranging species produce louder calls that travel farther, whereas species with quieter calls or specialized habitat use may be less reliably recorded.
3. **Data variability** — Many factors influence recording quality and bat activity, including weather, prey abundance and distribution, and local habitat. Additionally, stationary detectors may record an individual bat multiple times, meaning stationary acoustic data can only inform relative activity and occupancy.
4. **Processing limitations** — Auto-ID software enables efficient processing of large datasets, but is known to have high misclassification rates for bats. Manual verification generally produces more accurate results, but has its own limitations, including significantly higher processing time and cost. We use a combination of both approaches, as detailed in the [Manual Verification](#) subsection of the Methods.

2.4 Statistical Models

Bat monitoring data can be analyzed using different statistical approaches, each with distinct assumptions and outputs. Activity trend models examine changes in relative acoustic activity over time, providing a straightforward measure of whether bat detections are increasing, decreasing, or remaining stable across the monitoring network.

Dynamic occupancy models take a different approach, estimating the probability that a species occupies a given site, while explicitly accounting for imperfect detection. Rather than treating a non-detection as an absence, these models distinguish between a species being truly absent versus simply going undetected, an important consideration given the variable detectability of bats discussed above. They also track colonization and extinction probabilities across sites over time, providing insight into range shifts and local population dynamics.

In 2025, both model types were run to examine differences in results and assess the complementary information each approach provides.

3 Methods

3.1 Data Collection

3.1.1 Stationary Surveys

Two to four stationary detectors were deployed across at least two quadrants per grid cell for a minimum of seven nights. Sites were selected to maximize species detection and recording quality across different habitat types (see Rae and Blejwas (2026) for details). Deployment occurred between mid-May and mid-July, after bats have emerged from hibernation but before young bats are volant. Sites and timing were kept consistent between years to minimize variability in conditions affecting bat activity.

3.1.2 Transect Surveys

Where possible, driving transects were conducted following Loeb et al. (2015). Bats were recorded along the road by driving at ~20 mph with a microphone mounted vertically inside a magnetic pvc pipe holder on the vehicle's roof. Transects were 18-27 miles long and were surveyed twice during the stationary deployment period whenever possible.

3.1.3 Landscape and Weather Covariates

We considered the same covariates for the activity trend models as those previously established by Rae and Blejwas (2026). Temperature and relative humidity were recorded in each grid cell using onsite logging devices where possible; where loggers were unavailable, values were drawn from the nearest climate station. For sites in Southeast Alaska, station records were obtained using the Meteostat package in Python (Meteostat 2026), which aggregates weather station records from multiple agencies.

For sites in British Columbia, station records were obtained from the Pacific Climate Impacts Consortium (PCIC) Provincial Climate Data Set (PCDS), which aggregates and quality-controls observations from multiple provincial and federal networks. Additional weather covariates drawn from these station records included nightly mean wind speed (km/h) and nightly precipitation. Degree days above 10°C since January 1 were calculated from daily temperature records for each site-night.

Site-level covariates were derived from GIS (elevation, distance to water, water body type) or recorded by grid leaders during deployment (distance to clutter, percent clutter); estimates of percent clutter were confirmed using site photos. For sites with smaller ponds that did not appear on the GIS, we used the distance estimate recorded by the grid leader.

We explored additional factors influencing colonization and extinction in the dynamic occupancy models. For those models we included distance to the nearest road, and distance to timber harvest, calculated as the distance from each survey location to the closest road or harvest feature. Road and harvest features for British Columbia were derived from the unpublished Human Footprint Index (HFI) layer currently being developed by the Geospatial Centre within Biodiversity Pathways. Road features for Southeast Alaska were taken from the Alaska Department of

Transportation and Public Facilities ([AKDOT](#)) Roads layer and the US Forest Service Transportation layer. Harvest features for Southeast Alaska were taken from the US Forest Service Tongass_National_Forest_Past_Harvest_from_CoverType layer. For occupancy analyses, all continuous covariates were standardized by centering on their mean and dividing by their standard deviation; missing covariate values were imputed with the global mean prior to standardization.

3.1.4 Manual Verification

In 2024 and 2025 acoustic analysis mostly followed the standards set by WCSC in previous years (Rae and Blejwas 2026). This included running all files through two automatic classifiers, Kaleidoscope’s Bats of North America 5.4.0 classifier and Sonobat 3.0’s southeast Alaska classifier, to remove files that do not contain bat pulses and giving a preliminary bat species identification.

Manual vetting largely follows the protocol of Reichert et al. (2018) and previous analysis standards developed by WCSC (Rae and Blejwas 2026). The key difference in the 2024 and 2025 acoustic analysis compared to previous years is that not all automatic species identifications were manually verified. Instead, based on findings from Stratton et al. (2022), at least 50% of recordings were selected for manual verification.

3.1.4.1 Subsetting for manual verification

Given the project’s large scale, manually vetting all recordings is not feasible long-term. To address this, the NNW Bat Hub, in collaboration with WCSC and Brian Paterson, developed procedures to ensure that selecting a subset of recordings for manual review minimizes biases in analytical results. These standards prioritize the accurate identification of rare and difficult-to-recognize species to maintain the integrity of models and analyses.

To ensure data accuracy, manual verification was conducted based on the number of recordings that received an autoID tag at each site. For sites with 200 or fewer tagged recordings, all non-noise recordings were manually reviewed. For sites with more than 200 tagged recordings, verification depended on the proportion of recordings assigned a species label. All tags for rare species (comprising 10% of prior years’ identifications) and commonly misidentified species were fully vetted. Specifically, all Hoary Bat (*Lasiurus cinereus*) and Silver-haired Bat (*Lasiyonicteris noctivagans*) were reviewed due to the frequent misclassification of noise files that are tagged as these species. Additionally, recordings of common species were randomly selected for verification until 50% of total recordings were reviewed. If fewer than 50% of recordings had species tags, all classifier-assigned tags were manually verified, and “NoID” recordings were randomly selected until 50% of all recordings were reviewed.

Manual verification is complicated by the lack of information on species ranges within Southeast Alaska. We considered all species found in the region as potential candidates in all grid cells with the exception of Long-legged and Yuma Myotis. Long-legged Myotis calls are acoustically similar to those produced by the much more common Little Brown Myotis and auto ID software frequently misclassifies the calls of these two species. Unfortunately, although Little Brown Myotis produces diagnostic calls, Long-legged Myotis does not. Therefore, in grid cells without independent confirmation of the presence of Long-legged Myotis, we downgraded all auto IDs of that species to a Little Brown Myotis/Long-legged Myotis couplet. Where that species has been confirmed present, we retained the species identification unless diagnostic characteristics of Little Brown Myotis were

present. Calls of California Myotis also lack diagnostic characteristics and are acoustically similar to Yuma Myotis. However, to date Yuma Myotis has been documented only in the southern portion of Southeast Alaska (Olson and Gunderson 2014) and prior to 2024 we chose to exclude it as a potential species except for the Ketchikan and Prince of Wales grids. Beginning in 2024, we started including it in the species set for all grids in order to be able to detect any expansion of its range within Southeast Alaska.

3.2 Statistical Analyses

3.2.1 Activity Trends

We conducted species-specific trend analyses using negative binomial GLMM models as described in Rae and Blejwas (2026). We restricted the analysis to data from 2019 onward, due to the low number of grid cells that had been sampled in the years prior (2015 and 2017). We further limited the analysis to grid cells with at least five years of sampling, since estimating a temporal trend requires enough repeat visits per cell to distinguish a directional signal from year-to-year variation.

All models followed the same basic structure, modelling the total number of detected calls as a negative binomial response with log link:

```
Total detections ~ YearS + log(Sample Nights) + (1|Grid Cell ID) + (1|Quadrant)
+ species specific covariates
```

Where total detections is the total number of detected calls of a given species at a site in a given year. YearS is the standardized survey year (years since 2019) and provides the estimate of the temporal activity trend. log(Sample Nights) is the natural logarithm of the number of detector-nights and accounts for variation in sampling effort among site-years. The random intercepts (1|Grid Cell ID) and (1|Quadrant) allow baseline activity to vary among grid cells and among quadrants within them, respectively.

For each species, we performed model selection by fitting the base model augmented with every combination of up to four covariates, excluding combinations that would introduce collinearity. This produced 387 candidate models per species, from which we selected the model with the lowest AIC as the final covariate set Table 1.

Table 1: Top-performing stationary models selected via AIC-based stepwise selection. Covariates are described as, Distance to Clutter = distance (m) to nearest object; Percent Clutter = estimated percentage of space within a 30m radius obstructed by objects; Water Nearby = detector within 100m of water; WaterDist = distance (m) to nearest water feature; Nightly Temperature = nightly air temperature (°C); RH = nightly relative humidity (%); Nightly Mean Wind Speed = mean nightly wind speed (m/s); Nightly Precipitation = total nightly precipitation (mm); jNight = Julian date (day of year) of the survey night.

Species	Top Performing Model Covariates
Hoary Bat	
Silver-haired Bat	Percent clutter, Water nearby, Nightly min RH

California Myotis	Percent clutter, Nightly precipitation (mm)
Long-eared Myotis	Distance to clutter (m), Nightly max temp (°C), Nightly min RH, Julian night
Little Brown Myotis	Percent clutter, Distance to water (m), Nightly precipitation (mm), Nightly mean wind speed (m/s)
Long-legged Myotis	
Yuma Myotis	

Post-hoc tests were run on the final models to ensure proper model fit. Species were excluded from the analysis if model fit was not appropriate due to issues including under-dispersion or standard errors that were more than 3 times the estimate. Model results were also excluded if there was less than 150 recordings across all sample years.

3.2.1.1 Use of Automated Identification Results in Stationary Trend Analyses

For stationary trend analyses, species identifications came from the Kaelidoscope Pro AutoID software, not from manual verification. Manual review was used only to remove non-bat noise. Specifically, files that manual review flagged as noise were excluded, but where manual review suggested a different species than the AutoID label, the AutoID classification was retained.

This approach was first identified as necessary in 2023 (Rae and Lausen 2024), when analyses of manually verified data revealed a consistent positive trend across all species, this was interpreted as an artifact of reviewer learning over time rather than a true biological signal. As verifier experience increased across years, calls were increasingly assigned to specific species rather than left unidentified, producing an apparent increase in detections independent of actual bat activity. Because this observer effect is difficult to control for statistically, AutoID classifications are now used as the standard for all activity trend analyses, as automated classifiers apply consistent criteria across all years. While AutoID is subject to some misclassification, the error rate is consistent across years and does not confound detection of genuine temporal trends.

Trend analyses are presented only for species confirmed present through manual verification and for which models fully converged.

3.2.2 Transect Abundance Models

For the transect-based abundance trend models, we used only manually verified detections. When a detection was flagged as a species couplet, it was counted toward each species in the couplet. As above, we restricted the data to transect surveys from 2019 onward and to grid cells with at least five years of sampling.

All models followed the same basic structure, modelling the total number of detected calls as a negative binomial response with log link:

```
Total detections ~ YearS + log(transect length) + (1|Grid Cell ID)
+ species specific covariates
```

Where total detections is the total number of detected calls of a given species at a site in a given year. YearS is the standardized survey year (years since 2019) and provides the estimate of the temporal activity trend. $\log(\text{transect length})$ is the natural logarithm of the total length of the transect sampled and accounts for variation in sampling effort among transects. The random intercept (1|Grid Cell ID) allow baseline activity to vary among grid cells.

Covariates were limited to weather variables and transect length, and candidate models excluded any combination of collinear variables. We fit 59 candidate models per species and selected the one with the lowest AIC as the final covariate set Table 2.

Table 2: Top-performing transect models selected via AIC-based stepwise selection. Covariates are described as, Nightly Precipitation = total nightly precipitation (mm); jNight = Julian date (day of year) of the survey night.

Species	Top Performing Model Covariates
Hoary Bat	Nightly precipitation (mm), Julian night
Silver-haired Bat	
California Myotis	
Long-eared Myotis	
Little Brown Myotis	Nightly precipitation (mm)
Long-legged Myotis	
Yuma Myotis	

After selection, we ran a series of post-hoc checks to confirm model fit. We treated a species' result as reliable only when these tests agreed that the year effect reflected a genuine regional trend rather than an artefact of sparse counts or a single influential cell.

3.2.3 Dynamic Occupancy

We fitted multi-season (dynamic) occupancy models (MacKenzie et al. 2003) to estimate trends in occupancy, colonization, and persistence while accounting for imperfect detection. This analysis used nightly acoustic detection/non-detection data from passive bat detectors deployed at sites in Alaska and British Columbia (BC). Pooling the two regions allowed the more extensive BC data to inform species-specific detection probabilities and covariate slopes, improving our ability to distinguish true absences from non-detections and thereby sharpening occupancy estimates where Alaska data were sparse. To account for differences in detectability between Alaska and BC, we included a fixed regional effect on initial occupancy, colonization, and persistence.

Initial occupancy (ψ) was modeled as a function of distance to water and elevation. Colonization (γ) and persistence (φ) were modeled as a function of distance to water, elevation, year-varying distance to harvest, and distance to road. Detection probability (p) was modeled as a function of percent clutter, nightly mean temperature, and Julian night. Detection probability also included a fixed Alaska offset on the logit scale, allowing detectability to differ between Alaska vs. BC due to differences in species classification, equipment, and survey conditions such as shorter nights at higher latitudes. The offset was estimated from repeat visits at Alaska sites and was assumed constant across years. Species with few detections in Alaska (<30) make separating occupancy and detection difficult, thus Alaska estimates for those species were removed.

Data were filtered to the 2017–2025 period, and analysis used an every-other-night subsampling protocol to reduce temporal autocorrelation between consecutive nights, using a maximum of 4 nights per site-year. Sites had to have a minimum of 2 nights of data within a given year and data from at least 2 years total. Models were run separately for all 7 bat species. After this filtering, a total of 15 sites in 7 grid cells Southeast Alaska, as well as 185 sites in 63 grid cells in British Columbia (BC) were used for these analyses.

Models were fit in a Bayesian framework using JAGS (Version 4.3.2) via the `jagsUI` package (Version 1.6.3) in `r` (Version 4.5.1). We ran 3 chains in parallel for 30,000 iterations each, discarding the first 15,000 as burn-in and thinning the remainder by 10, yielding 9,000 posterior samples for inference. Priors on all regression coefficients were normal with mean 0 and precision 0.368 (equivalent to a standard deviation of approximately 1.65), and the standard deviation of the region-level random effect was assigned a $\text{Uniform}(0, 5)$ prior. Convergence was assessed using $\hat{R}$ diagnostics, with values below 1.1 taken to indicate convergence. Results are summarized using posterior means and 95% Bayesian credible intervals. We calculated an overall occupancy trend metric () as the ratio of estimated occupancy in the final year to that in the initial year; values of > 1 with 95% credible intervals entirely above 1 indicate an increase in occupancy, values < 1 with intervals entirely below 1 indicate a decline, and intervals overlapping 1 indicate an uncertain or stable.

4 Results

In 2024 all 9 grid cells were monitored, and in 2025 a total of 6 grid sites were sampled in SE Alaska and 3 newly established sites were sampled in the Anchorage area (Figure 2).

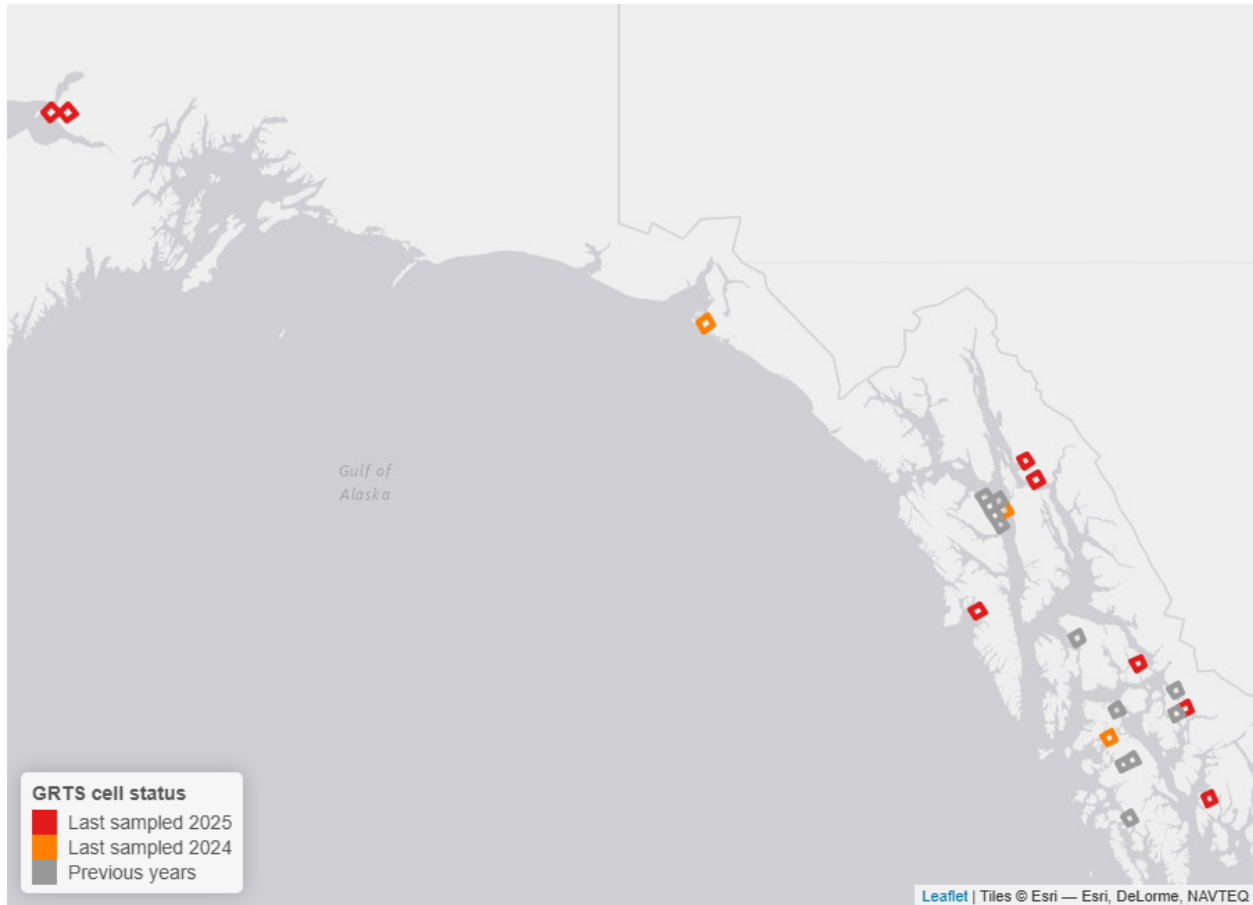


Figure 2: Acoustic bat monitoring sites in Alaska in 2025. NABat grid cells monitored (with two or more stationary sites) in 2025 appear in red and those monitored in 2024 in orange. Cells monitored in previous years but not in 2024 or 2025 appear in grey.

Total acoustic activity varied substantially across sites and years, with Little Brown Myotis dominating call volume at every site in nearly every year. Overall detection totals were highest at the northern sites: Yakutat produced the single largest site-year total (~17,500 calls in 2024), and Juneau recorded consistently high volumes (~12,000 in 2020 and ~13,500 in 2024). Southern and central sites generally recorded lower totals, with Petersburg (~9,000 in 2024) and POW (~7,500 in 2024) reaching comparable peaks in single years, while Wrangell and Hoonah remained low throughout (generally under ~2,000 calls per year). Most sites recorded substantially higher activity in 2024 than in any other year.

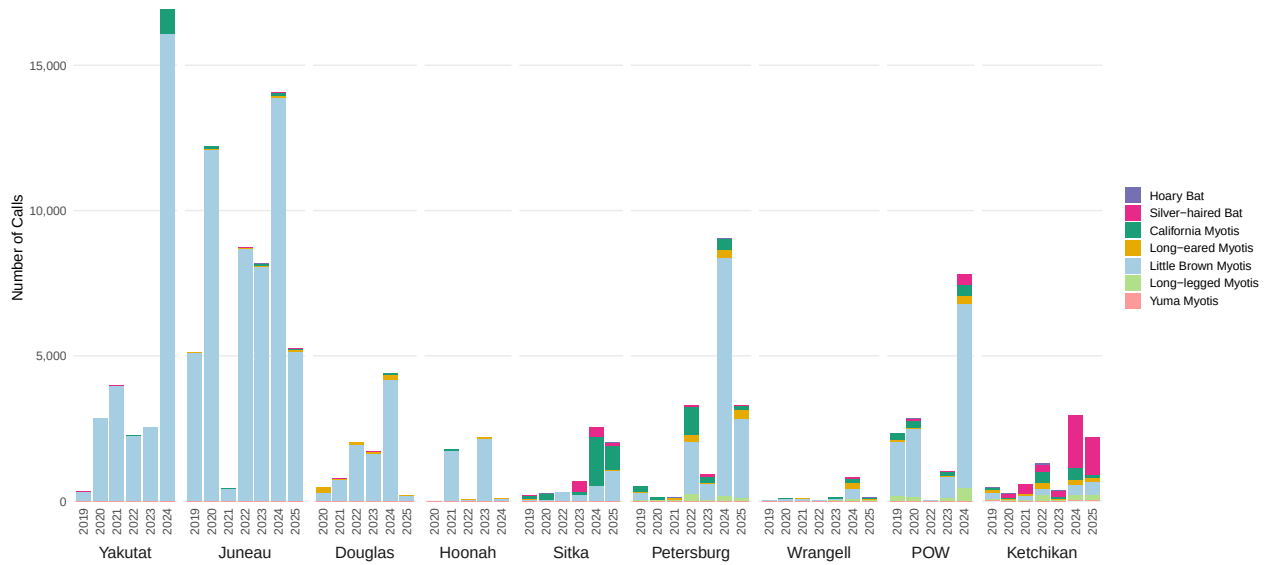


Figure 3: Relative activity of bat species detected during NABat monitoring in 9 grid cells in Southeast Alaska, arranged north to south, during 2019-2025

4.1 Activity and Abundance Trends

We estimated yearly trends in nightly acoustic activity for seven species across the stationary detector and mobile transect survey streams. Five species produced reportable stationary-detector models; two (Long-legged Myotis, Yuma Myotis) could not be reliably fit and are not reported (Table 3). For transect abundance trends we only report on the results of Silver-haired bat and Little Brown bat since these were the only two species where modelling results were reliable.

Silver-haired Bat showed the clearest and most consistent signal, with activity increasing significantly over the monitoring period in both the stationary ($\beta = 0.48 \pm 0.17$, $p < 0.001$) and transect ($\beta = 0.40 \pm 0.20$, $p = 0.001$) datasets, the only species for which a significant trend was recovered independently in both survey types.

California Myotis showed an increase in activity from the stationary model ($\beta = 0.14 \pm 0.13$, $p = 0.036$). Long-eared Myotis ($\beta = 0.06 \pm 0.13$, $p = 0.35$) and Little Brown Myotis (stationary $\beta = 0.08 \pm 0.14$, $p = 0.24$; transect $\beta = 0.00 \pm 0.12$, $p = 0.94$) showed no detectable trend, with confidence intervals spanning zero in both reportable models.

Hoary Bat, Long-legged Myotis and Yuma Myotis are not reported: Hoary bat and Long-legged Myotis fell below the detection threshold required for trend estimation, and the Yuma Myotis model failed to converge.

Table 3: Trend estimates from NABat stationary and transect monitoring in southeast Alaska, 2019–2025, adjusted for covariates. Estimates are on the natural-log scale. Stationary and transect models were fit separately as independent pipelines; blank cells indicate species that did not meet the criteria for reporting in that framework. Significant results ($p < 0.05$) are shown in bold.

Species	Stationary Models		Transect Models	
	Trend Estimate $\pm$ 95%CI	P-value	Trend Estimate $\pm$ 95%CI	P-value
Hoary Bat				
Silver-haired Bat	0.48 ± 0.17	<0.001	0.40 ± 0.20	<0.001
California Myotis	0.14 ± 0.13	0.036		
Long-eared Myotis	0.06 ± 0.13	0.352		
Little Brown Myotis	0.08 ± 0.14	0.242	0.00 ± 0.12	0.937
Long-legged Myotis				
Yuma Myotis				

4.1.1 Covariate Effects

Beyond the year term used to estimate activity trends, each species’ best-supported model retained one or more weather or temporal covariates, and stationary models could additionally retain site-level descriptors (Table 4). Because transect surveys traverse heterogeneous terrain within a cell, site-specific covariates such as distance to clutter, percent clutter, and distance to water were eligible only in the stationary models.

Covariates describing moisture in the air were among the most frequently retained across species, entering models either as relative humidity or as nightly precipitation. Retained weather terms were consistently negative in sign, indicating lower nightly activity under wetter conditions, a pattern consistent with reduced foraging during precipitation and elevated humidity.

Table 4: Covariate effects from the best-fitting (lowest-AIC) negative binomial generalized linear mixed model for each species’ stationary and transect acoustic detector data. Estimates are fixed-effect coefficients on the log link ($\pm 1.96 \times$ standard error, an approximate 95% confidence interval). Only species with a reportable stationary trend model are included.

Covariate	Stationary Models		Transect Models	
	Estimate $\pm$ 95% CI	P-value	Estimate $\pm$ 95% CI	P-value
Silver-haired Bat				
Nightly min RH	-0.05 $\pm$ 0.05	0.068		
Nightly precipitation (mm)			0.43 $\pm$ 0.34	0.014
Percent clutter	-0.02 $\pm$ 0.02	0.040		
Water nearby	1.34 $\pm$ 0.96	0.006		
Julian night			-0.09 $\pm$ 0.03	<0.001
Little Brown Myotis				
Nightly mean wind speed (m/s)	-0.31 $\pm$ 0.33	0.067		
Nightly precipitation (mm)	-0.20 $\pm$ 0.17	0.016	-0.21 $\pm$ 0.31	0.199
Percent clutter	-0.03 $\pm$ 0.02	<0.001		
Distance to water (m)	-0.00 $\pm$ 0.00	0.002		
Long-eared Myotis				
Distance to clutter (m)	-0.02 $\pm$ 0.02	0.004		
Nightly max temp ($^{\circ}$ C)	0.31 $\pm$ 0.21	0.005		
Nightly min RH	0.05 $\pm$ 0.04	0.014		
Julian night	-0.06 $\pm$ 0.04	0.002		
California Myotis				
Nightly precipitation (mm)	-0.33 $\pm$ 0.24	0.006		
Percent clutter	0.04 $\pm$ 0.01	<0.001		

4.2 Dynamic Occupancy

Dynamic occupancy models converged for all species. Due to the low number of Hoary bat (LACI = 3) and Yuma Myotis (MYU=6) detections across the monitoring period all Hoary bat and Yuma Myotis results were removed from this report. Two species showed increasing occupancy over the 2017–2025 period (Table 5). Little brown Myotis occupancy increased by an estimated 3.0 percentage points per year (95% CI: 0.2–7.0), corresponding to a net increase of 22.7 percentage points between 2017 and 2025. The net change was imprecise (95% CI: -13 to $+73$), but occupancy was higher in 2025 than in 2017 in 78% of posterior draws. Long-legged Myotis occupancy increased by 3.7 percentage points per year (95% CI: 0.8–5.6), with a net increase of 23.9 percentage points (95% CI: -6.7 to $+40$). Although this credible interval also overlapped zero, the probability that occupancy was higher in 2025 than in 2017 was 95%, providing the strongest evidence of an increase among the species assessed.

Trends for silver-haired bat, California Myotis and long-eared Myotis were uncertain, with 95% credible intervals for both the trend slope and net change overlapping zero (Table 5). Silver-haired bat showed a possible decline in occupancy (trend slope -0.007 ; -0.21) with an estimated 23% probability that occupancy was higher in 2025 than in 2017. However, part of this apparent decline

may reflect the model adjusting from an uncertain starting value rather than a true loss. Long-eared Myotis showed a possible increase (slope 0.028; net change 0.18), while California Myotis showed no clear direction. Continued monitoring, particularly more consistent coverage of Alaska sites, is needed to resolve trends for these species.

Table 5: Estimated occupancy trends for seven bat species in Southeast Alaska (2017–2025) from a dynamic occupancy model. Trend slope is the average annual change in the proportion of sites occupied (e.g., 0.03 = +3 percentage points per year). is the net change in the proportion of sites occupied from 2017 to 2025 (e.g., 0.23 = 23 percentage points higher in 2025). Values are posterior means with 95% credible intervals. P(increase) is the proportion of posterior draws in which occupancy was higher in 2025 than in 2017; values near 0.5 indicate no clear direction, and values near 1 or 0 indicate stronger evidence of an increase or decline. Bold rows indicate trend slopes whose 95% credible interval excludes zero.

Species	Trend Slope (95% CI)	(95% CI)	P(Increase)
Hoary Bat	-0.00942 (-0.06444,0.05614)	-0.1863 (-0.8667,0.6)	0.3031
Silver-haired Bat	-0.0072 (-0.05222,0.04556)	-0.2088 (-0.6667,0.3333)	0.2291
California Myotis	0.01986 (-0.02444,0.07889)	-0.0608 (-0.5333,0.5333)	0.3504
Long-eared Myotis	0.02776 (-0.01111,0.07333)	0.1756 (-0.2667,0.6667)	0.6664
Little Brown Myotis	0.0304 (0.00222,0.07)	0.2273 (-0.1333,0.7333)	0.7809
Long-legged Myotis	0.03716 (0.00778,0.05556)	0.2393 (-0.0667,0.4)	0.9458
Yuma Myotis	0.00702 (-0.04667,0.07333)	-0.0049 (-0.6667,0.7333)	0.4367

4.2.1 Covariate Effects

Covariate effects varied by species, but temperature was the most consistent detection covariate, with detection probability increasing on warmer nights for most species (Figure 4). Clutter also influenced detection probability, although the direction and strength of this effect were species-specific. Detection probability differed between Alaska and British Columbia for hoary bat and yuma Myotis, which is expected given that both species are rarely detected in the areas sampled. Detection probability for silver-haired bat and California Myotis was also lower in Alaska than in British Columbia, but the reason for this difference is unknown.

Elevation and distance to water were the covariates most commonly associated with initial occupancy, colonization, and persistence. Although not always significant, colonization and persistence generally declined with increasing distance to water across species. Distance to water was also a significant predictor of initial occupancy for California Myotis and of persistence for little brown Myotis and long-legged Myotis. For yuma Myotis, persistence increased with distance from timber harvest, while for hoary bat, colonization decreased with distance from roads, suggesting this species may use roads as travel corridors when moving into new sites.

More detailed species results can be found on [Appendix A](#).

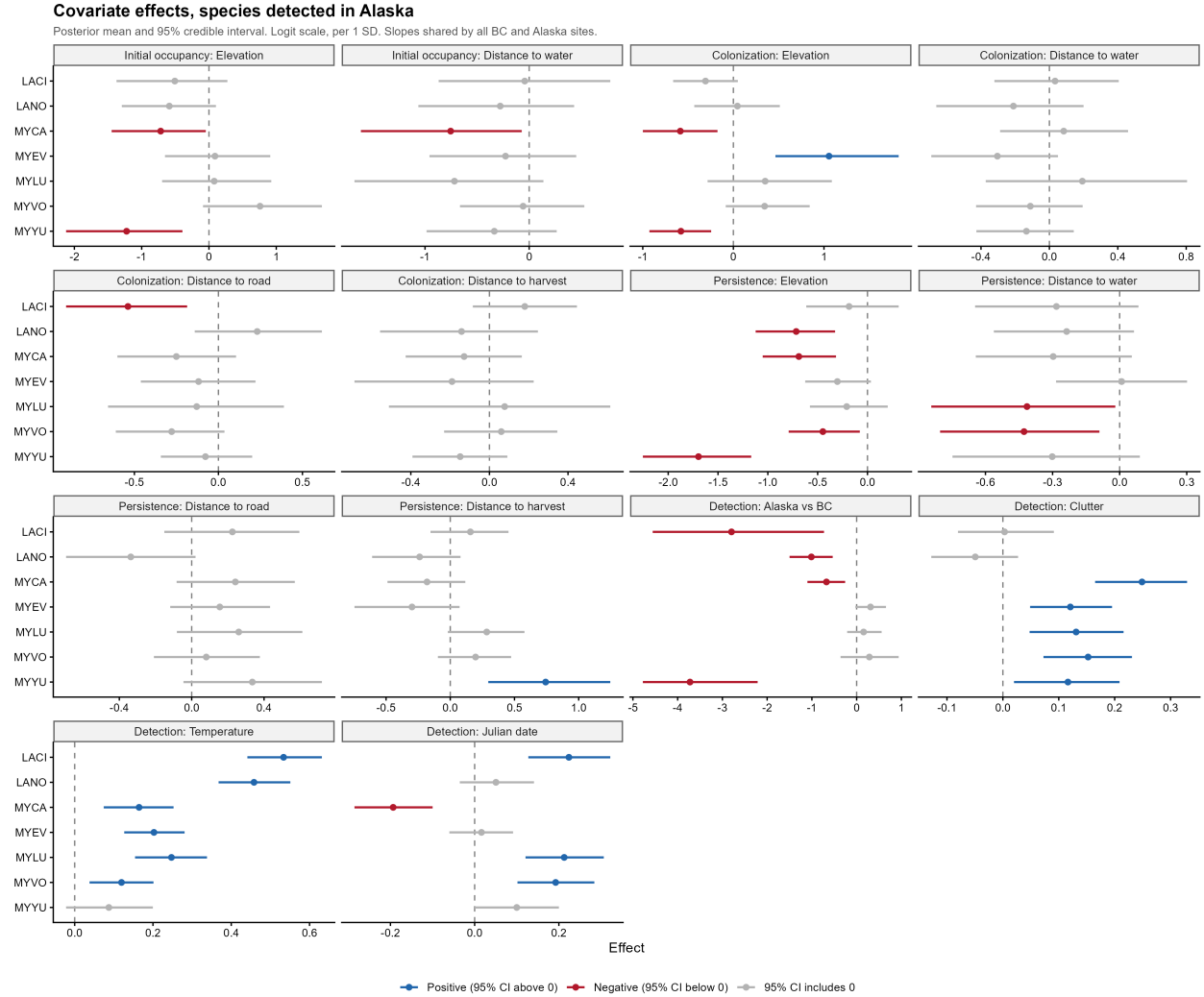


Figure 4: Covariate effects on initial occupancy, colonization, persistence, and detection probability for each species, modelled using data from Alaska and British Columbia. Effects whose 95% confidence intervals do not overlap zero are shown in colour (blue for positive, red for negative); all others are shown in grey.

5 Discussion

5.1 Interpreting Alaska estimates vs. BC's contributions

Alaska's 15 sites were too few to support a model of their own, so they were analysed together with the 185 BC sites, with Alaska included as a separate region through fixed regional effects. Some estimates are therefore specific to Alaska: initial occupancy, average colonization and persistence rates, and the occupancy state of each Alaska site in each year. Other estimates are shared across regions and cannot be separated from BC: the effects of elevation, distance to water, distance to roads and distance to harvest on occupancy dynamics; year-to-year variation in colonisation, persistence and detection; and the effects of clutter, temperature and date of detection. Because Alaska contributes fewer than 10% of the sites, these shared terms are effectively estimated from BC data and applied to Alaska. This assumes that habitat relationships, and the timing of good and poor years, are broadly similar across regions.

5.2 Trend Results

Overall, Alaska bat populations do not show evidence of decline, and several species show potential increases: the Silver-haired Bat (LANO) in activity, and most *Myotis* species in occupancy.

The positive activity trends for the California *Myotis* (MYCA) and LANO should be interpreted with caution (Table 3). Activity counts are noisy, and it can be difficult to distinguish a true trend from a random increase driven by year effects. For this reason it is prudent to examine trends across many years; a trend that persists across years is more likely to reflect a real change. The 2023 Alaska report identified a negative and non-significant trend for MYCA (Rae and Blejwas 2026), so although the estimate is significant in the present analysis, additional years of data are needed to confirm it.

In contrast, Rae and Blejwas (2026) found a positive activity trend for LANO that was marginally non-significant ($p = 0.05$), and the continued positive trend observed in this report (stationary: 0.48 ± 0.17 $p < 0.001$, and transect 0.40 ± 0.20 $p < 0.001$) suggests it may represent a genuine trend rather than noise. Two additional results point the same way. The occupancy results from this report show that although the occupancy trend is not significant, it is positive (Table 5). The second independent study by Udell et al. (2025) has preliminary NABat species-distribution and trend models that estimate an increase in LANO occupancy across Alaska (mean = 0.23, 95% CI [0.07, 0.39]). The accumulation of estimates pointing consistently in the same direction, across different datasets and modeling approaches, gives us a solid point from which to conclude LANO is increasing in occupancy across southeast Alaska. This pattern also speaks to why we include occupancy analyses alongside activity trends in this report: activity counts are noisy and easily perturbed by year effects, so a metric that is less sensitive to fluctuations in activity volume offers a valuable complementary check.

Occupancy models require a stronger signal to detect change, because they consider only whether a cell is occupied rather than overall activity levels. As a result, when a trend is detected in occupancy we can be more confident that it reflects a real change. Our occupancy results indicate that all *Myotis* species except MYCA may be increasing in occupancy (Table 5; Appendix A), which could suggest that *Myotis* species are expanding in Southeast Alaska. It is worth noting that MYCA is the one *Myotis* species that does not show an increase in occupancy, in direct contrast to its

positive activity trend. This divergence reinforces the caution above: the MYCA activity estimate should not be treated as confirmed until additional years of data are available. With that caveat in mind, the broader occupancy signal is encouraging, particularly because these are the species most likely to be affected by white-nose syndrome (WNS).

One important consideration is the large pulse in activity observed across most sites in 2024. Sampling effort in 2024 was comparable to previous years and to 2025, so this spike most likely reflects year-to-year variation in bat activity rather than a change in survey design. This pulse currently registers as part of the trend in the models because of its magnitude, but additional years of data will help discern whether it represents a true trend or a short-lived effect of an atypical year.

5.3 Looking forward

The reduced survey effort in 2025 was the result of staffing cuts at the US Forest Service, and staffing levels are unlikely to improve in the near future. Maintaining sampling at the sites and grid cells with the most continuous data is vital to ensuring that we can continue to detect population changes, especially given the imminent threat of WNS. If the number of grids must be reduced, priority should be given to retaining those that have been sampled the longest.

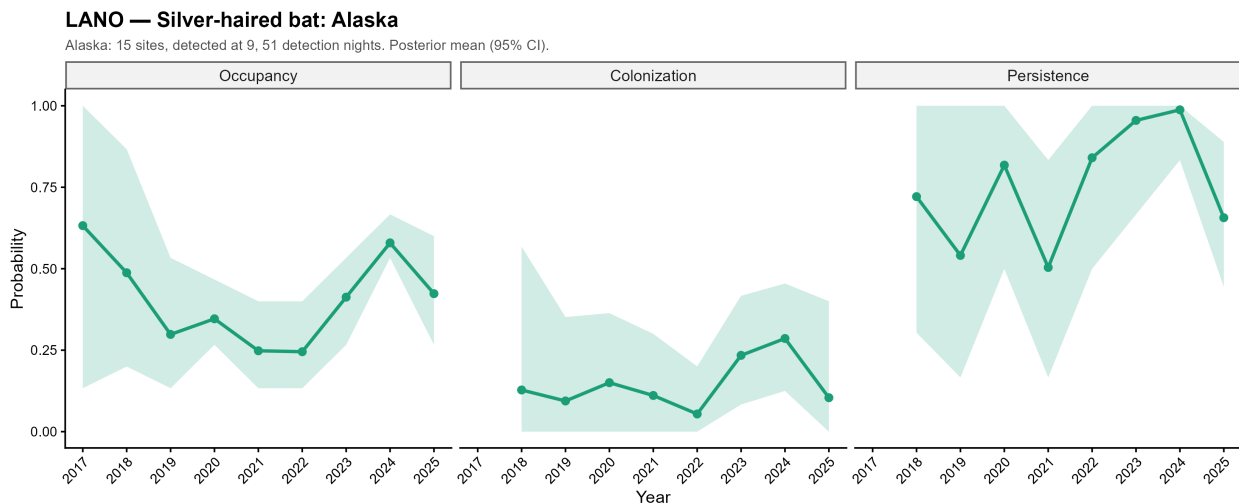
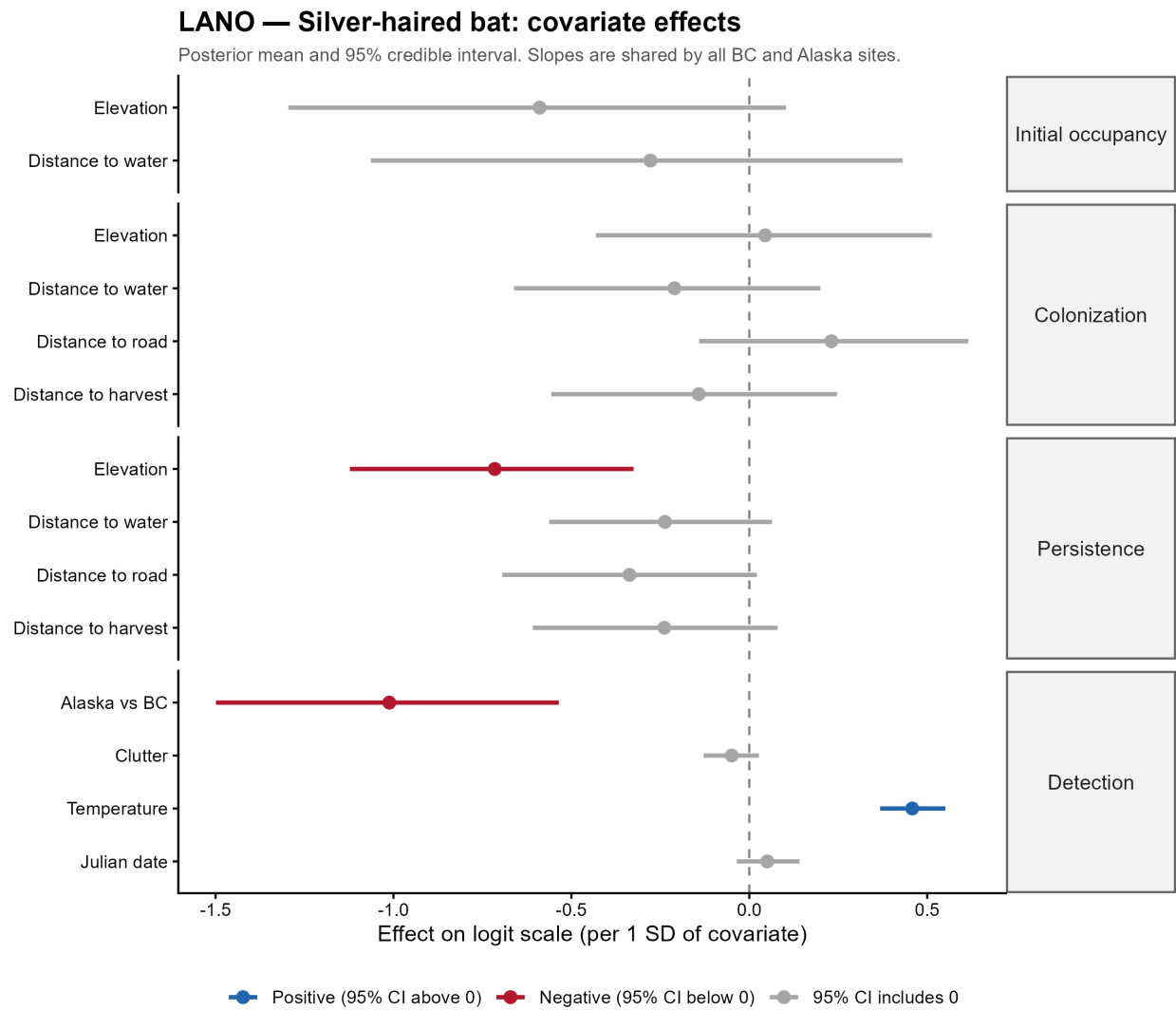
Collaboration with other agencies could also help sustain monitoring at existing or new sites. One approach currently being explored in British Columbia and Alberta is engaging industry partners, such as those in the forestry and wind energy sectors, in coordinated monitoring at their sites, and having them contribute both to the hub and to the broader effort. Such partnerships may be easier to establish where regulatory frameworks support them, for example by requiring that monitoring data collected at industry sites be submitted to NABat.

Another way to sustain monitoring is to look for ways to make the process more efficient. Grid leaders should continue to be encouraged to use Survey123 to streamline site deployment; in 2024 and 2025 this platform was not used consistently across all grids. Using it also helps ensure that metadata are consistent and complete. In 2026 we can also examine efficiencies in weather data collection, specifically evaluating whether continued use of on-site temperature and humidity loggers yields results comparable to weather data drawn from the nearest weather stations.

As species ranges continue to expand, a review of the data aimed at identifying potential new species would be prudent, particularly the Big Brown Bat (EPFU). Because this species produces calls very similar to those of LANO, bioacoustics could be used for preliminary screening to find potential sites worth exploring further. Once these sites are identified, capture surveys can be conducted, as capturing bats will ultimately be the only way to confirm a range expansion.

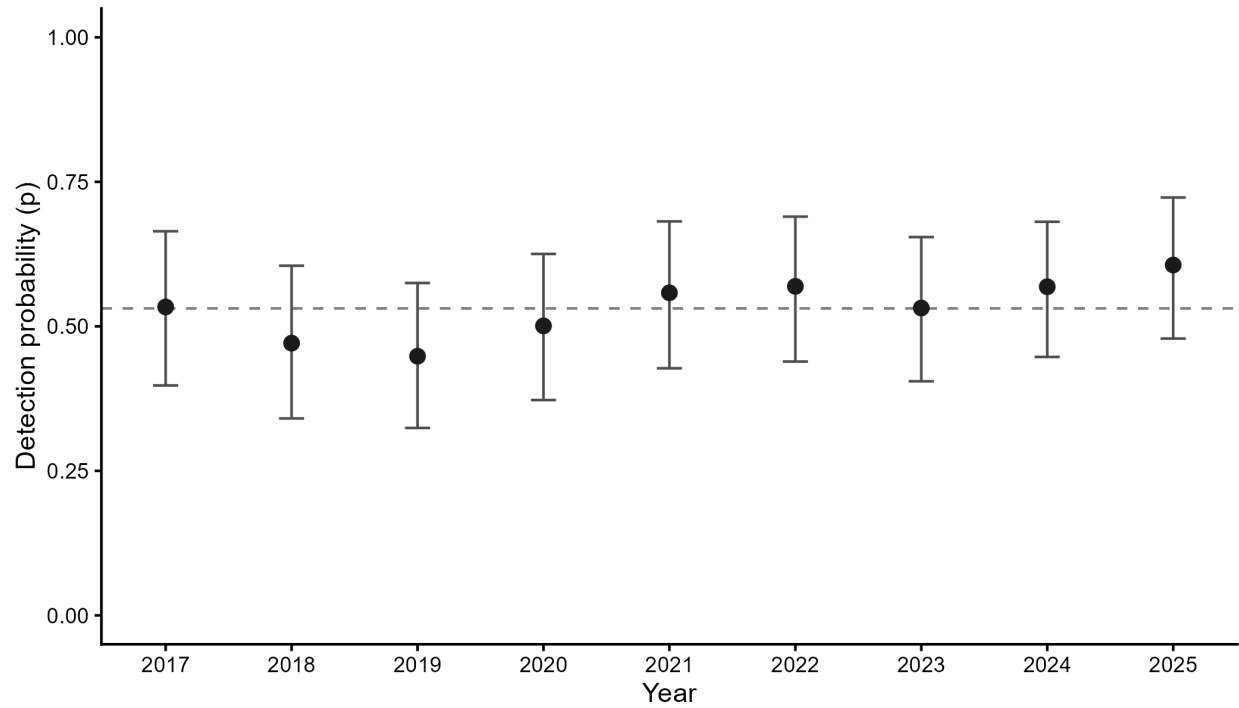
6 Appendix A

6.1 Silver-haired Bat (*Lasionycteris noctivagans*)

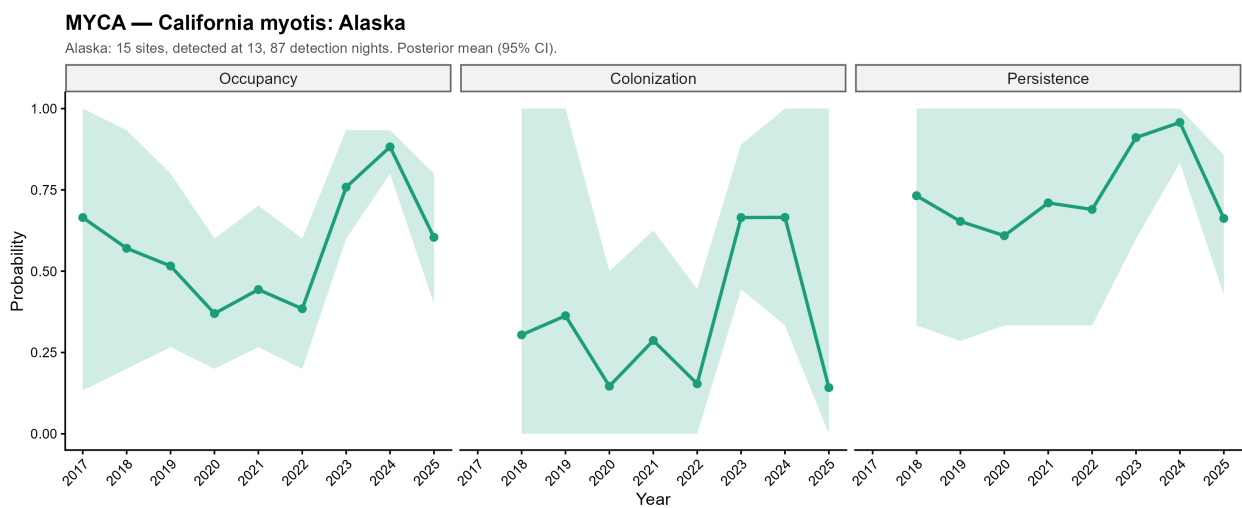
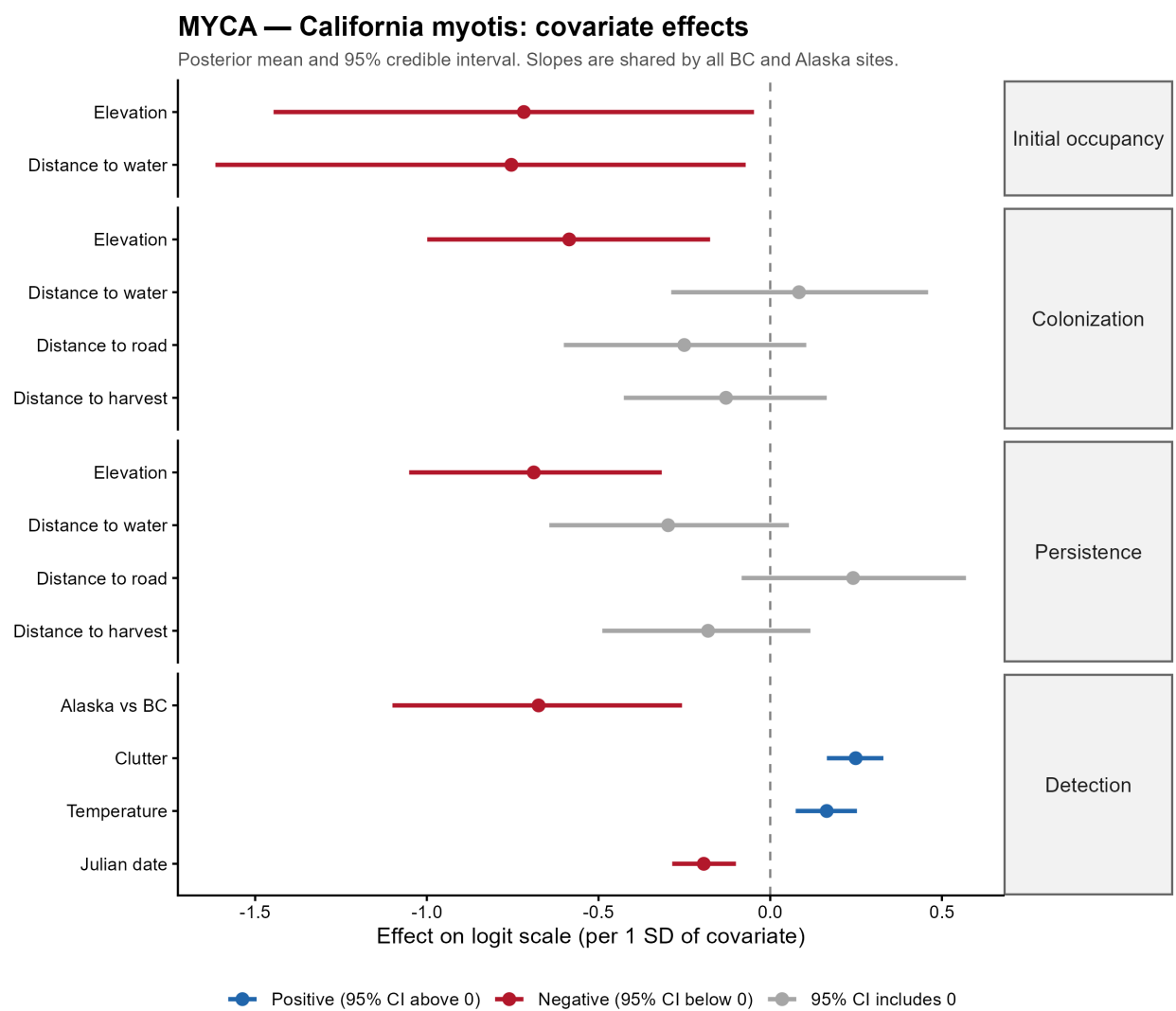


LANO — Silver-haired bat: detection probability by year, Alaska

Per-visit detection at Alaska sites, at average covariate values (95% CI). Dashed line = overall mean.

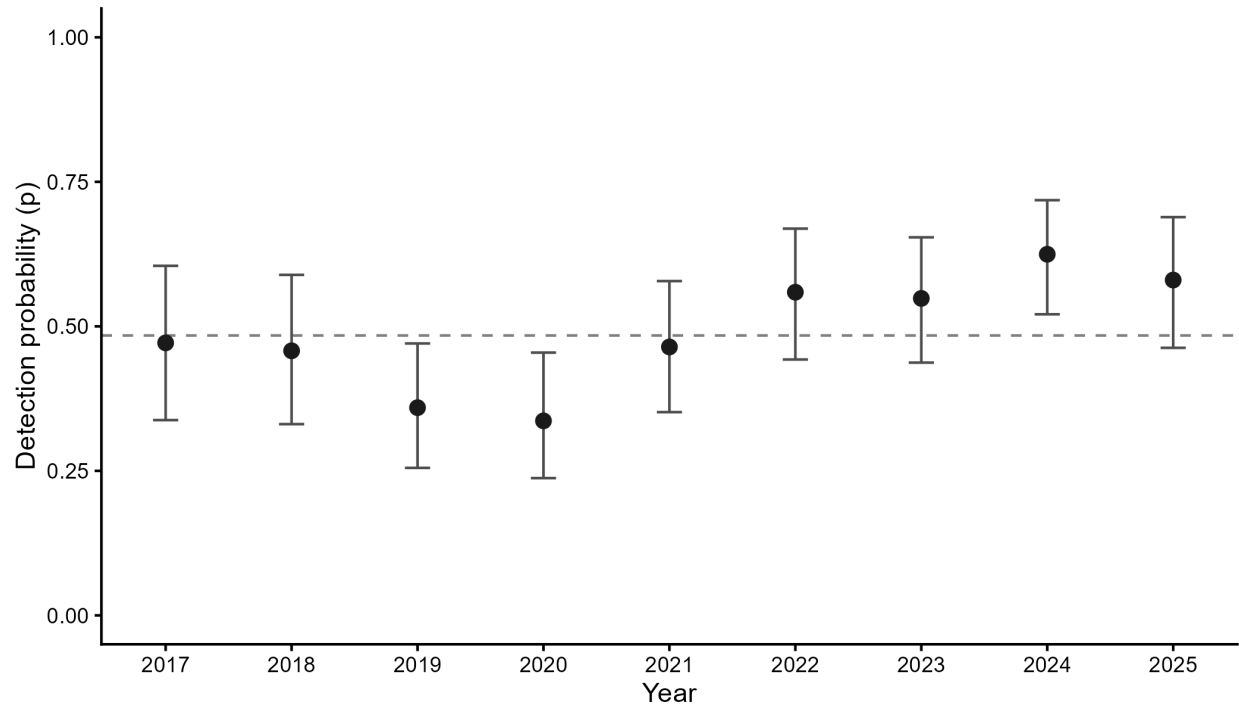


6.2 California Myotis (*Myotis californicus*)

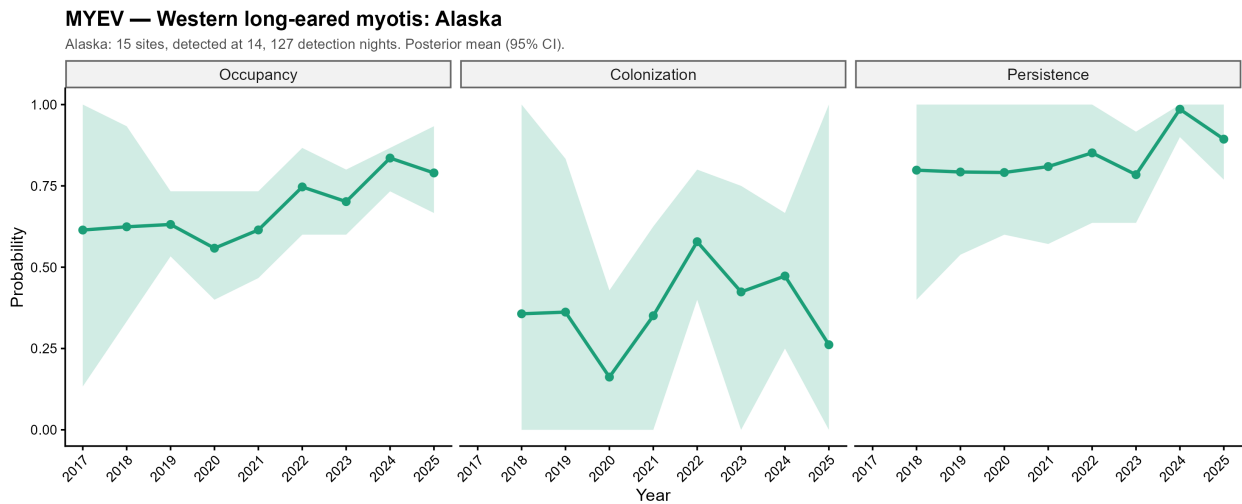
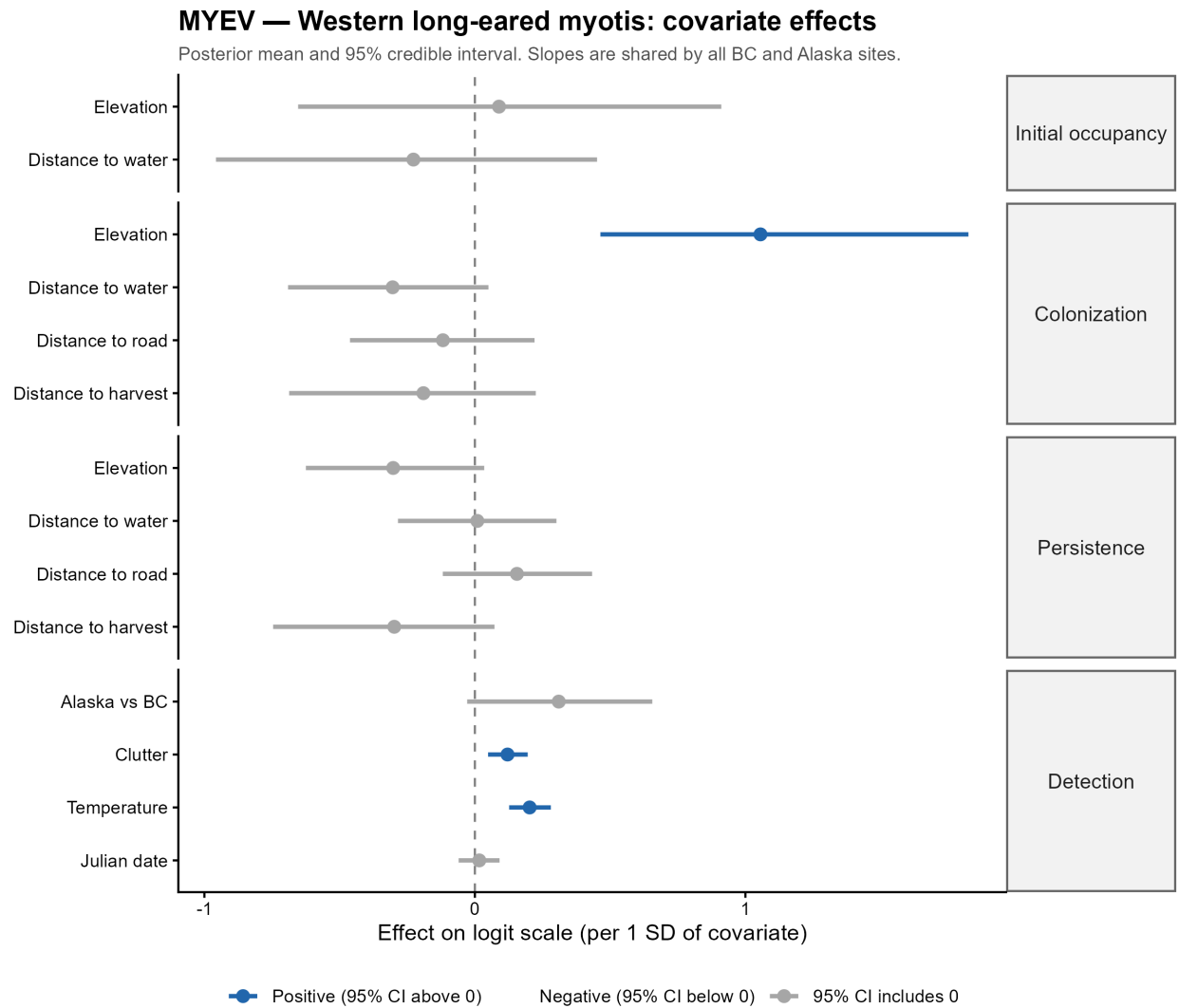


MYCA — California myotis: detection probability by year, Alaska

Per-visit detection at Alaska sites, at average covariate values (95% CI). Dashed line = overall mean.

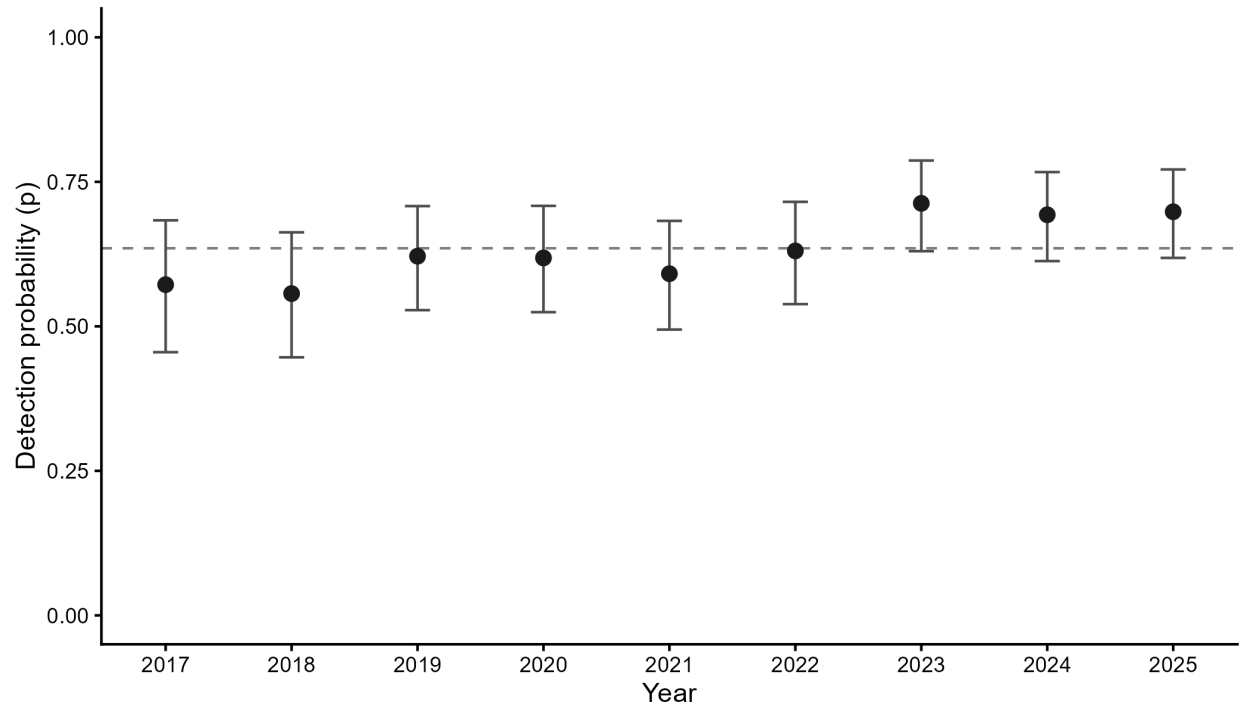


6.3 Long-eared Myotis (*Myotis evotis*)

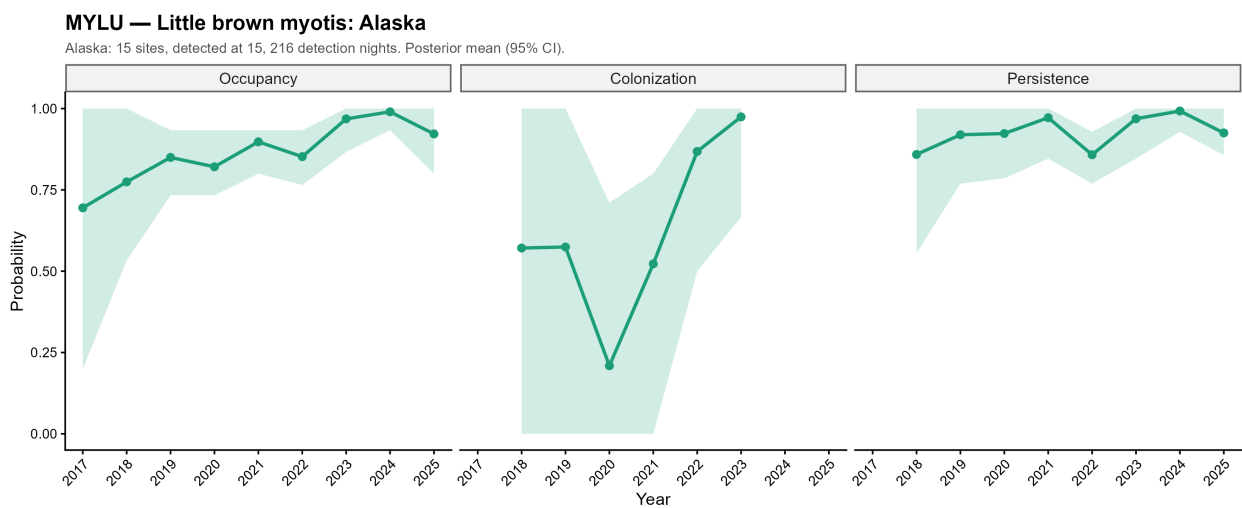
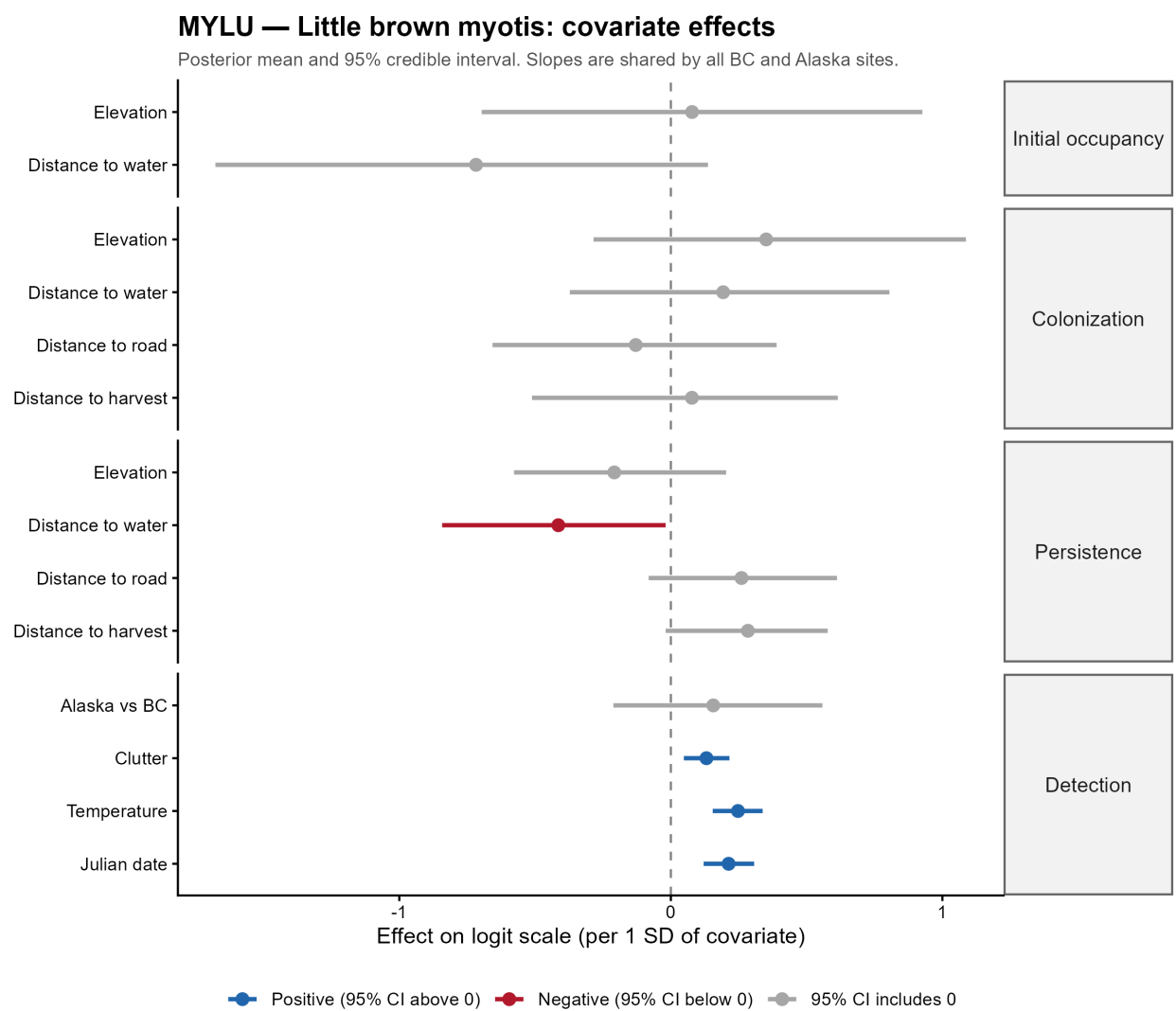


MYEV — Western long-eared myotis: detection probability by year, Alaska

Per-visit detection at Alaska sites, at average covariate values (95% CI). Dashed line = overall mean.

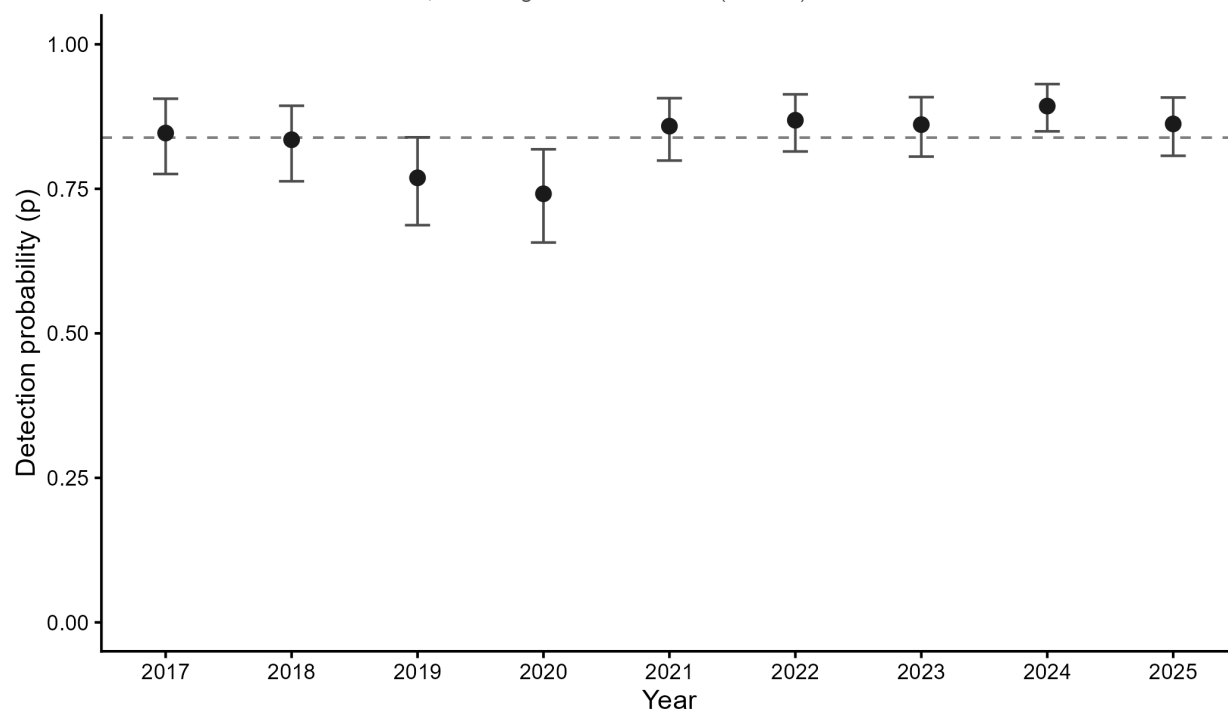


6.4 Little Brown Myotis (*Myotis lucifugus*)

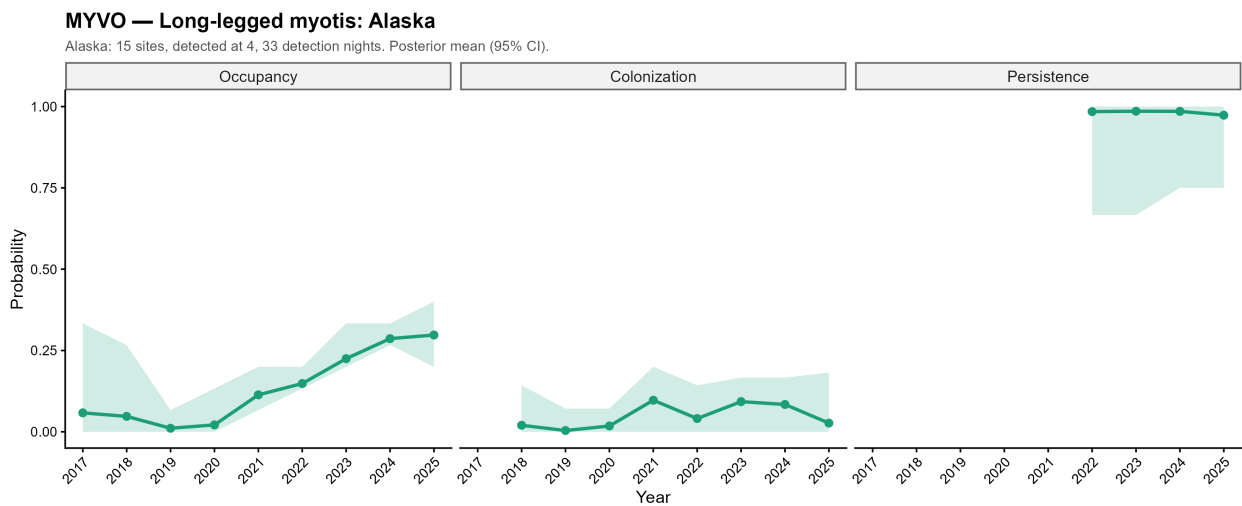
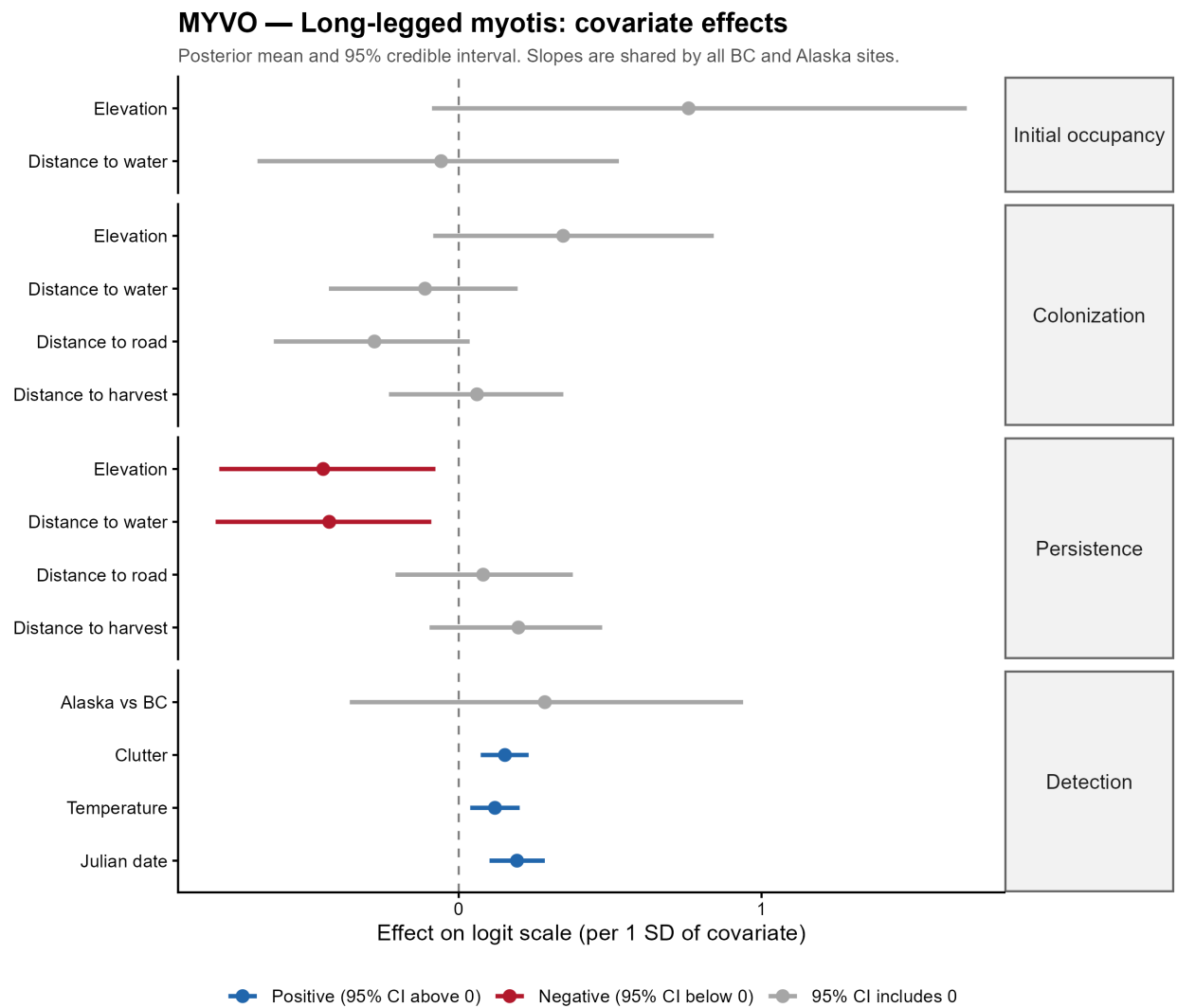


MYLU — Little brown myotis: detection probability by year, Alaska

Per-visit detection at Alaska sites, at average covariate values (95% CI). Dashed line = overall mean.

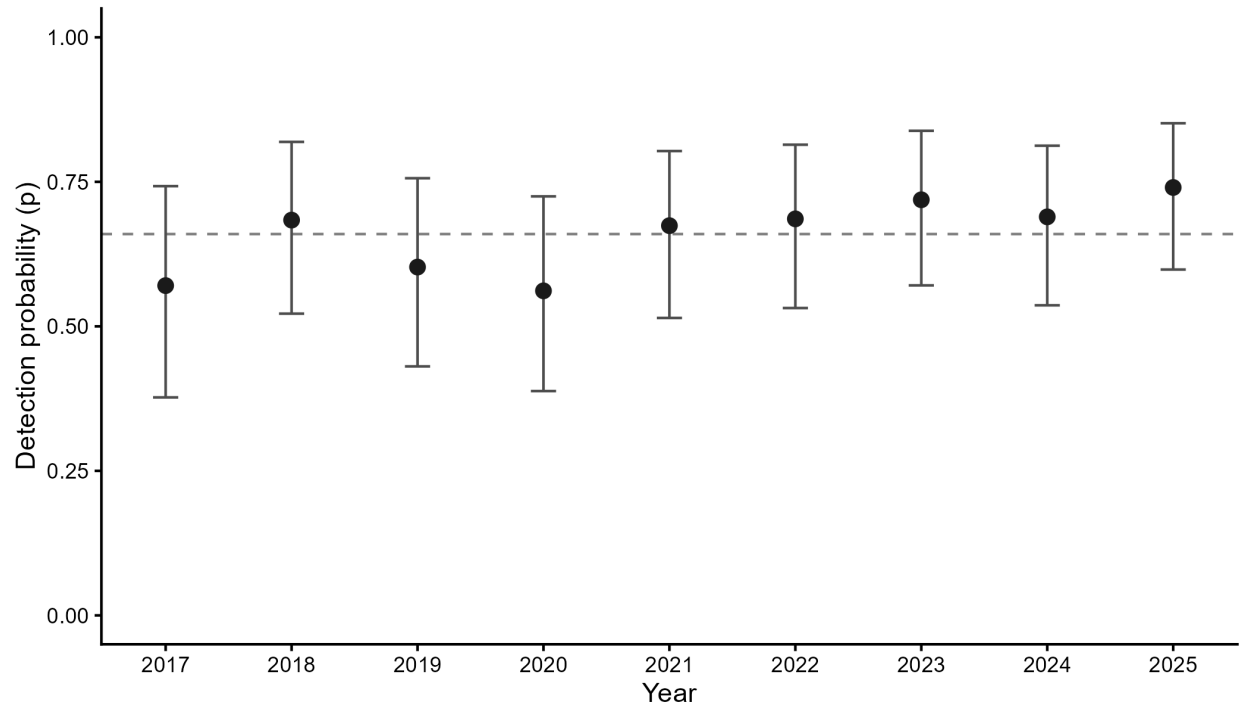


6.5 Long-legged Myotis (*Myotis volans*)



MYVO — Long-legged myotis: detection probability by year, Alaska

Per-visit detection at Alaska sites, at average covariate values (95% CI). Dashed line = overall mean.

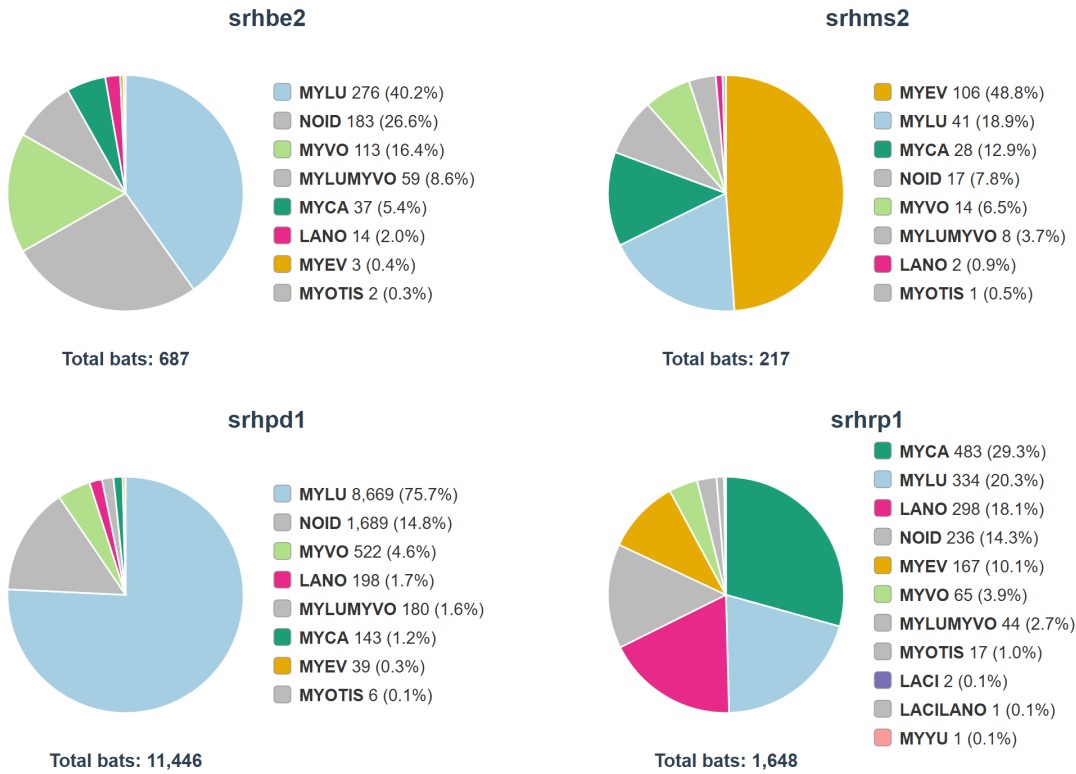


7 Appendix B

Species detection summaries by GRTS cell. For each cell, 2025 plots appear first, with the 2024 equivalents below where the cell was sampled in 2024.

Grid cell 206 - POW

2024



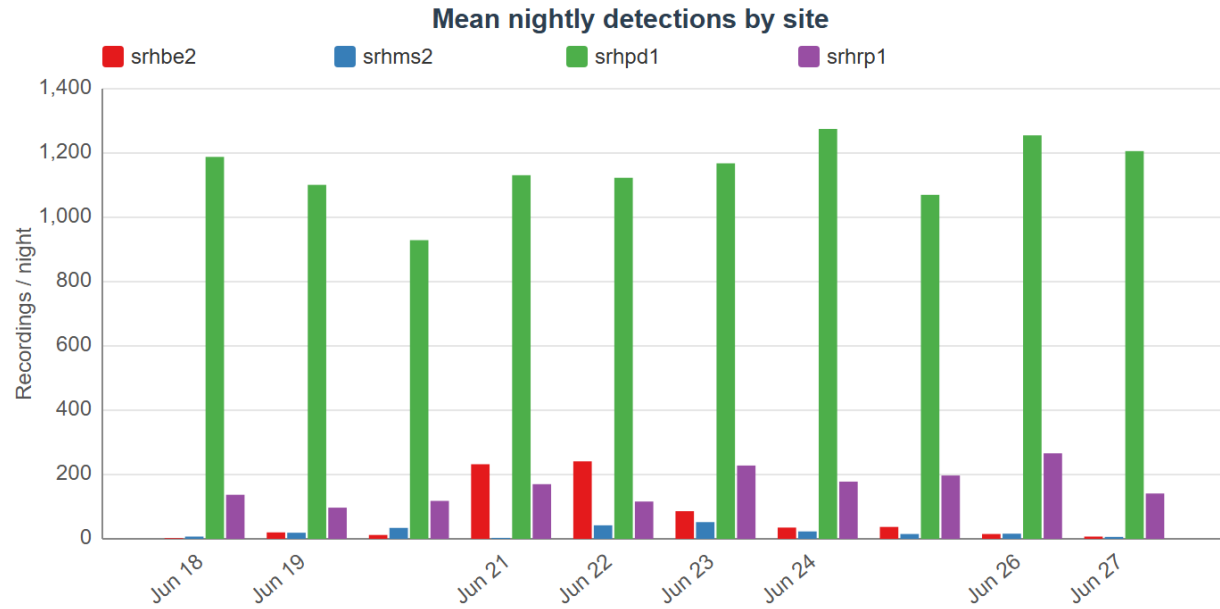


Figure 5: Nightly activity by site (2025 top, 2024 below), cell 206 - POW

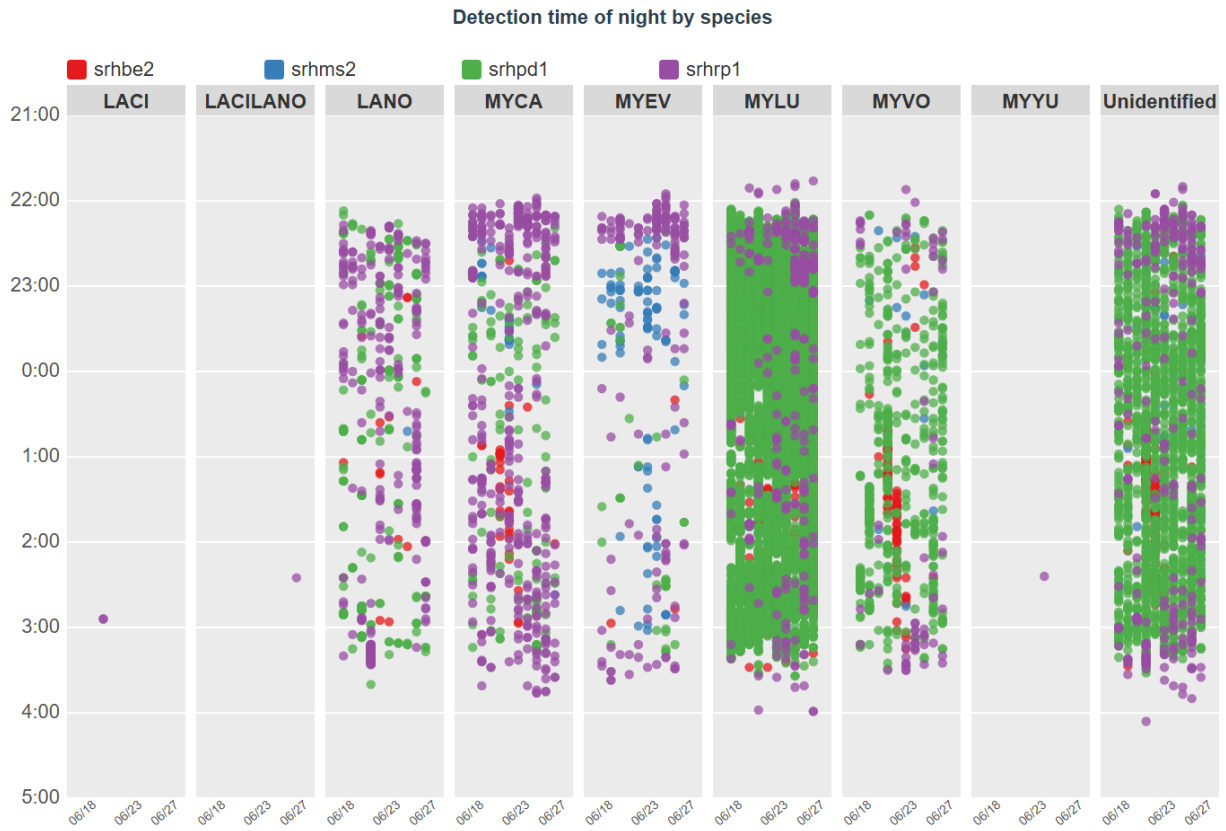
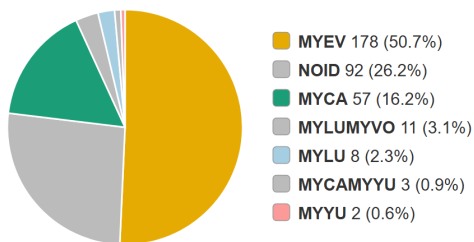


Figure 6: Detection times by species and site (2025 top, 2024 below), cell 206 - POW

Grid cell 5006 - Petersburg

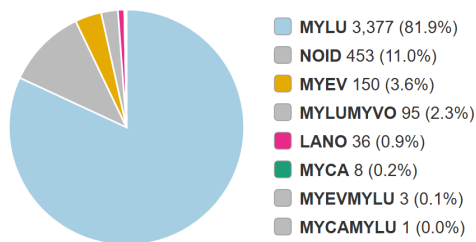
2025

tlfr2



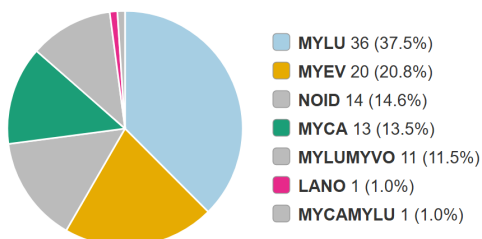
Total bats: 351

tlk2



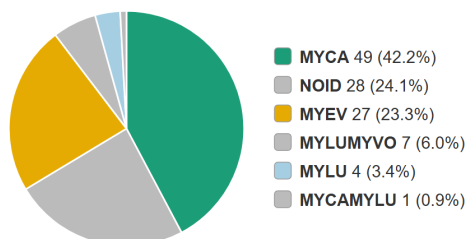
Total bats: 4,123

tlms1



Total bats: 96

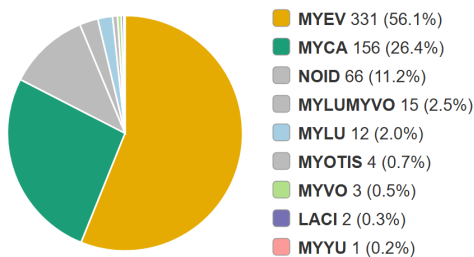
tlrp2



Total bats: 116

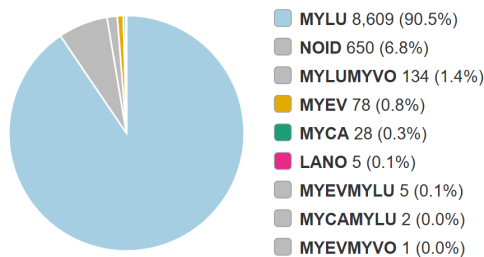
2024

tlfr2



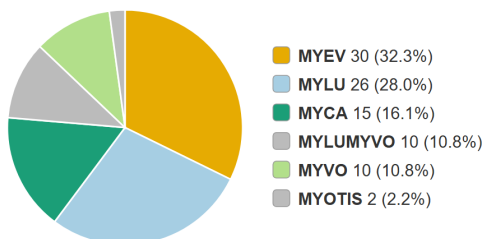
Total bats: 590

tlk2



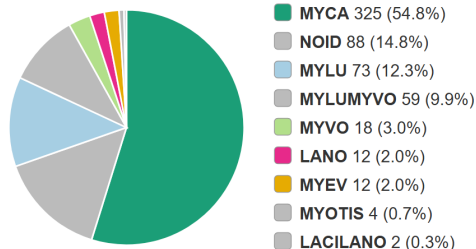
Total bats: 9,512

tlms1



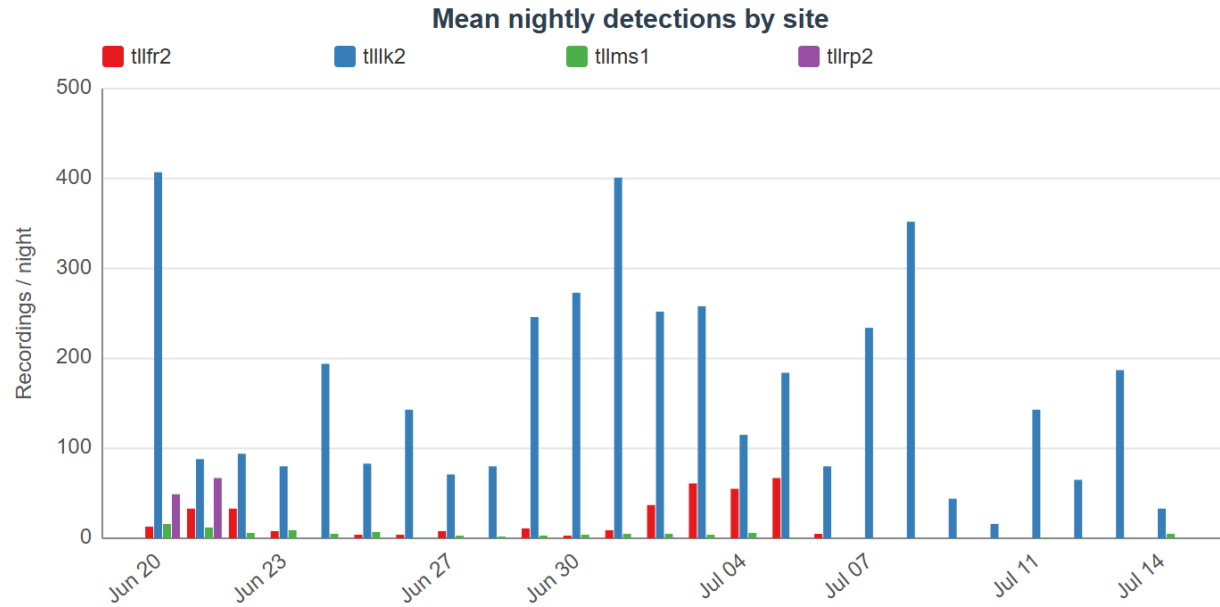
Total bats: 93

tlrp2



Total bats: 593

2025



2024

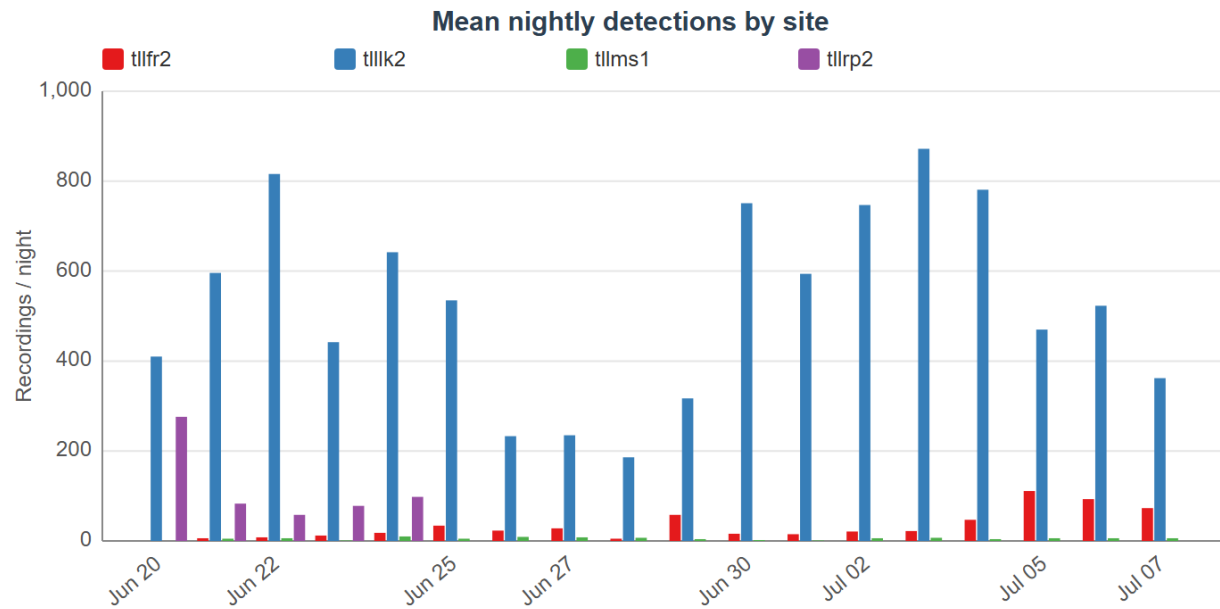


Figure 7: Nightly activity by site (2025 top, 2024 below), cell 5006 - Petersburg

2025



2024

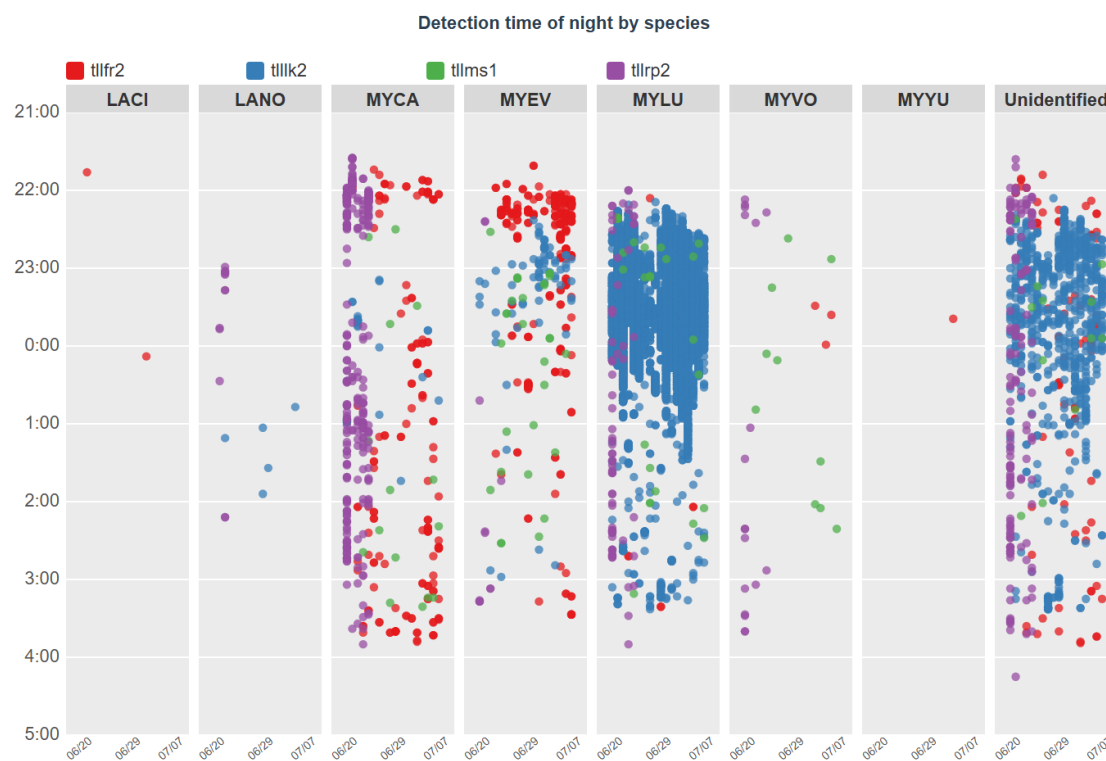
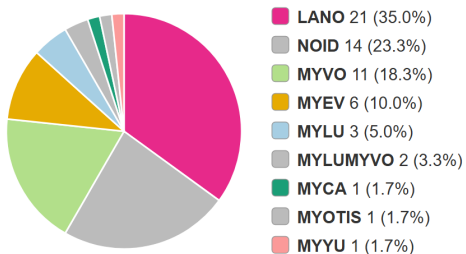


Figure 8: Detection times by species and site (2025 top, 2024 below), cell 5006 - Petersburg

Grid cell 18318 - Wrangell

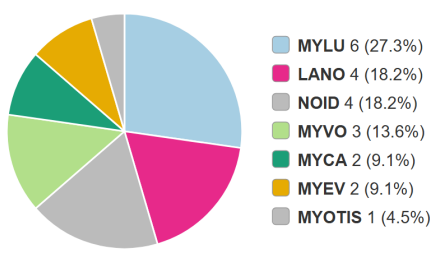
2025

foofr2



Total bats: 60

fooms1

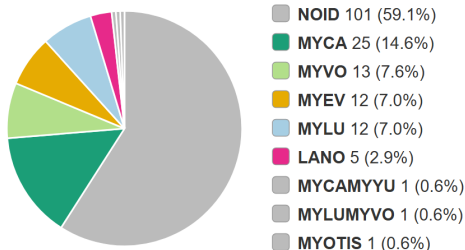


Total bats: 22

foopd1

No detections

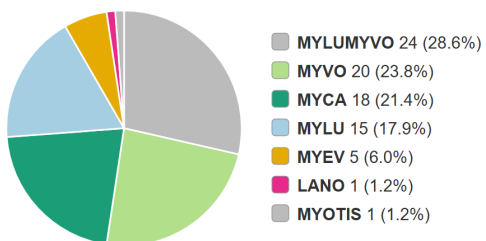
foorp1



Total bats: 171

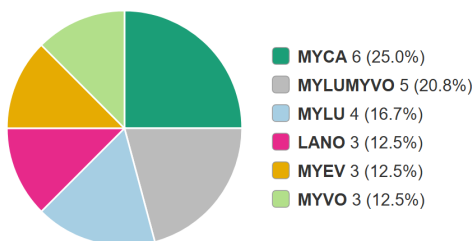
2024

foofr2



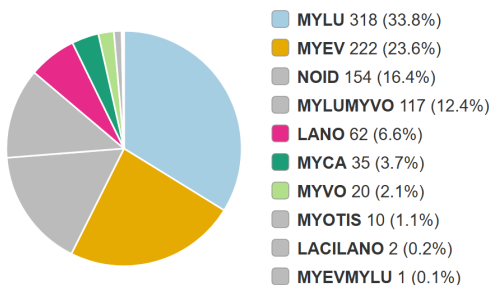
Total bats: 84

fooms1



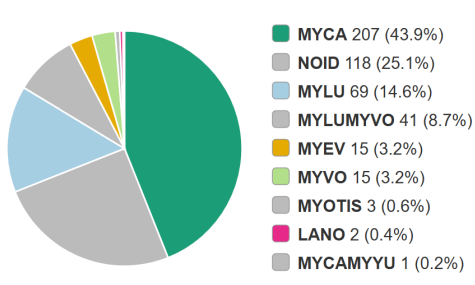
Total bats: 24

foopd1



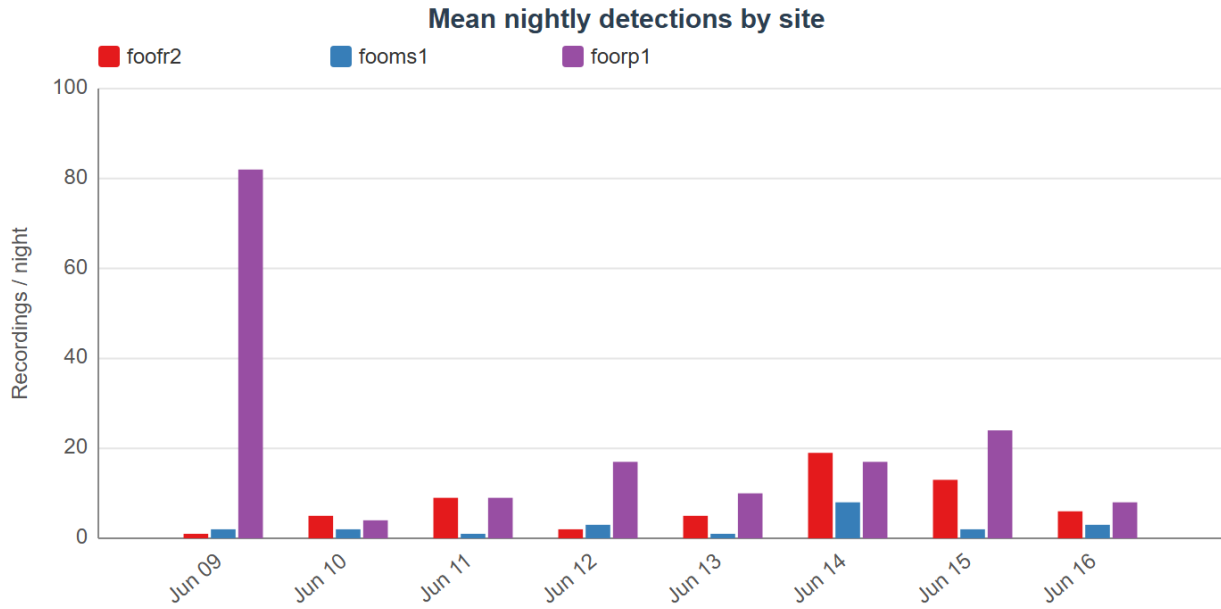
Total bats: 941

foorp1



Total bats: 471

2025



2024

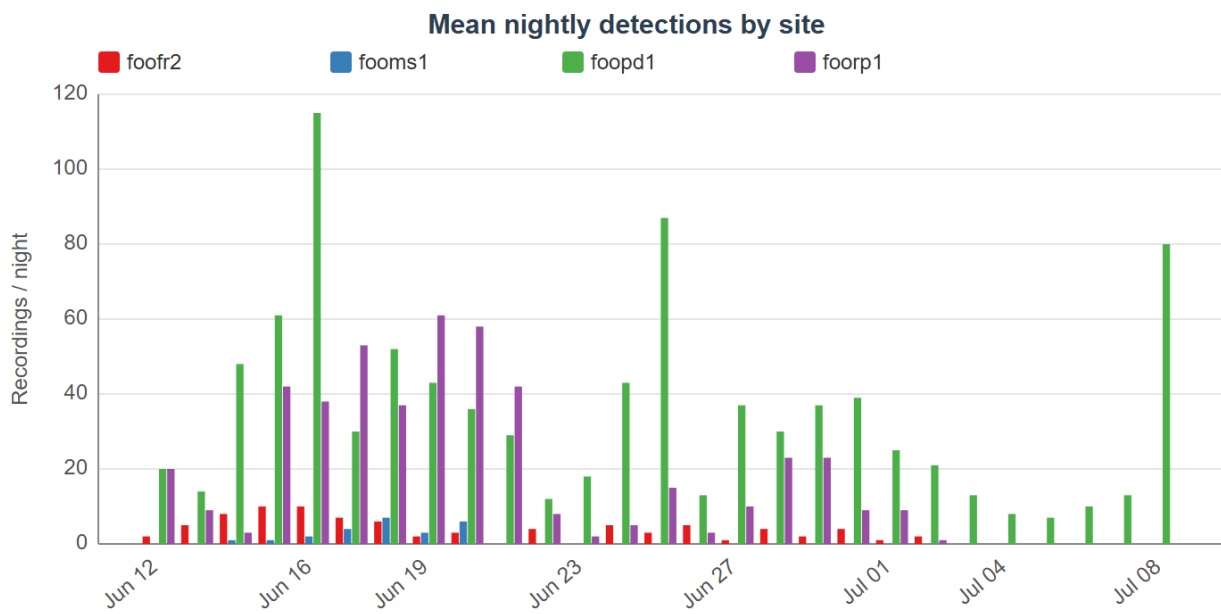
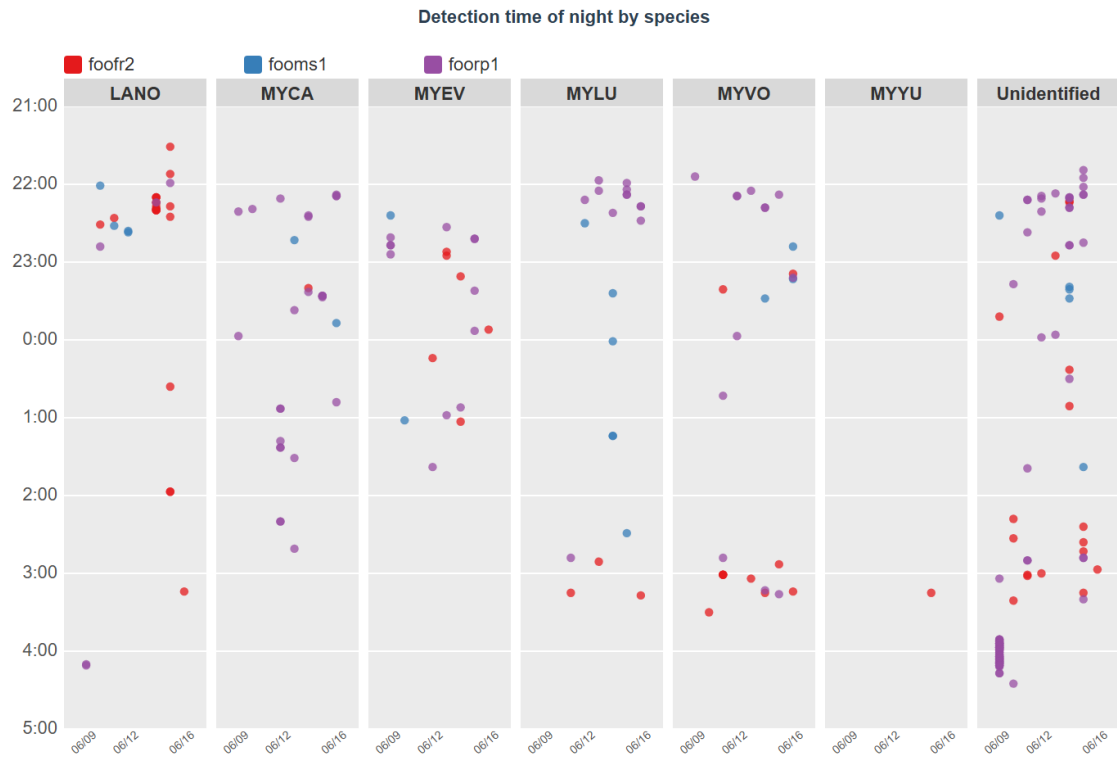


Figure 9: Nightly activity by site (2025 top, 2024 below), cell 18318 - Wrangell

2025



2024

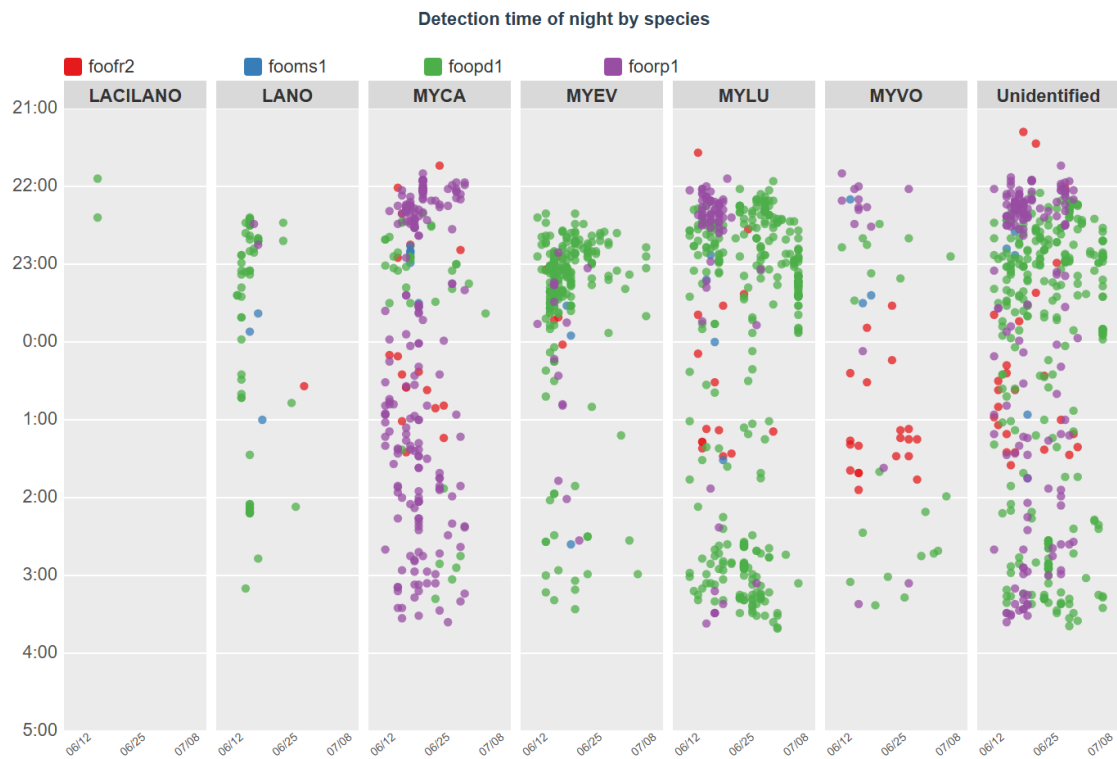
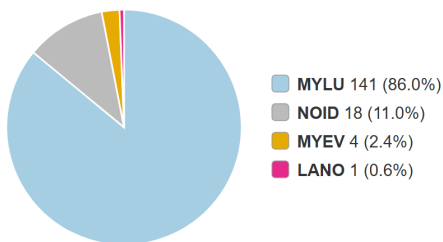


Figure 10: Detection times by species and site (2025 top, 2024 below), cell 18318 - Wrangell

Grid cell 31438 - Douglas

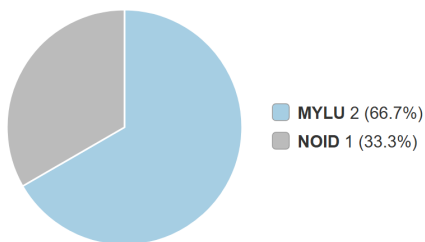
2025

doube2



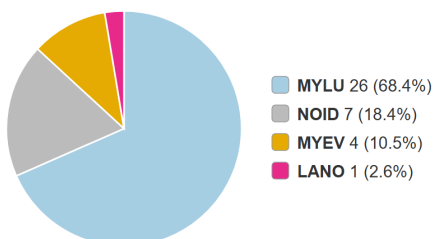
Total bats: 164

doufr1



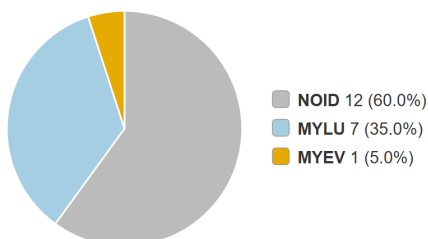
Total bats: 3

doums1



Total bats: 38

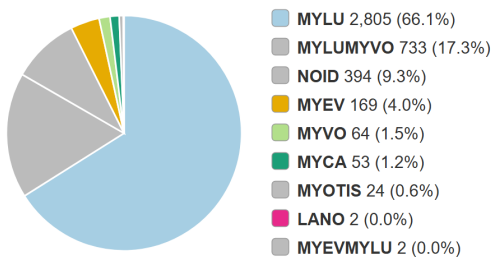
dourp1



Total bats: 20

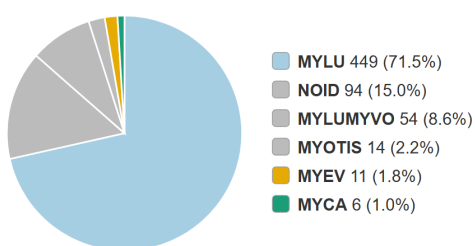
2024

doube2



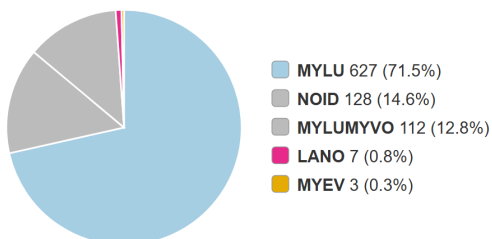
Total bats: 4,246

doufr1



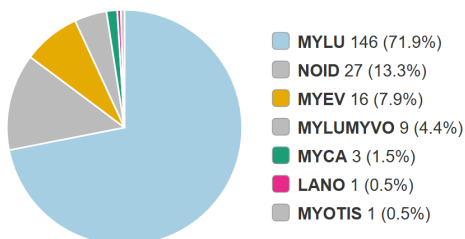
Total bats: 628

doums1



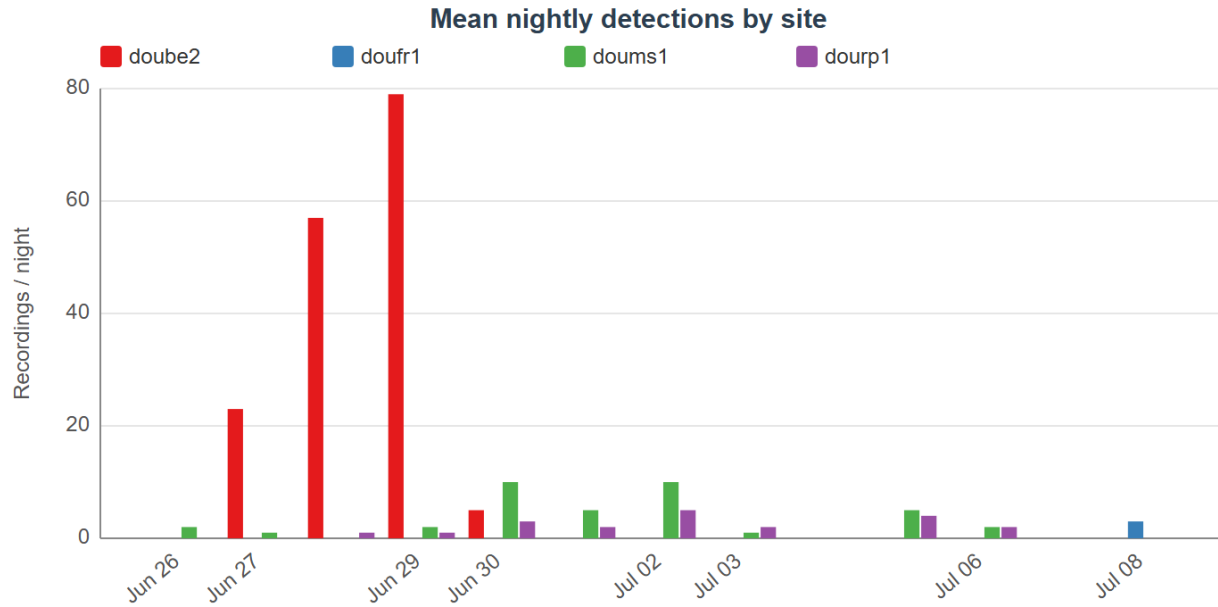
Total bats: 877

dourp1



Total bats: 203

2025



2024

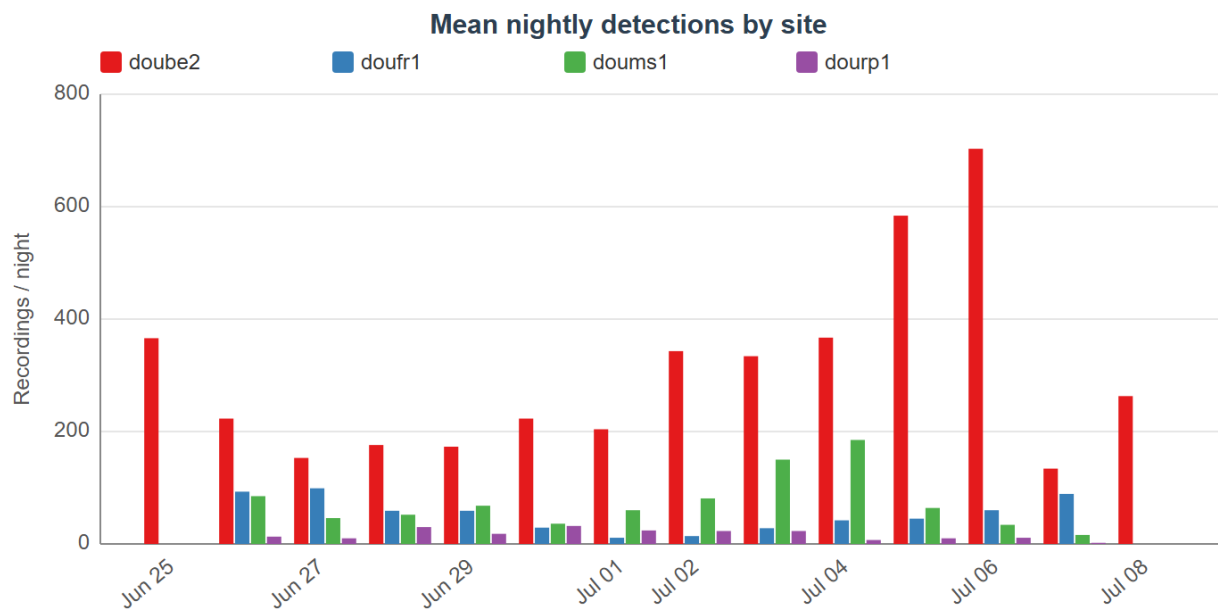
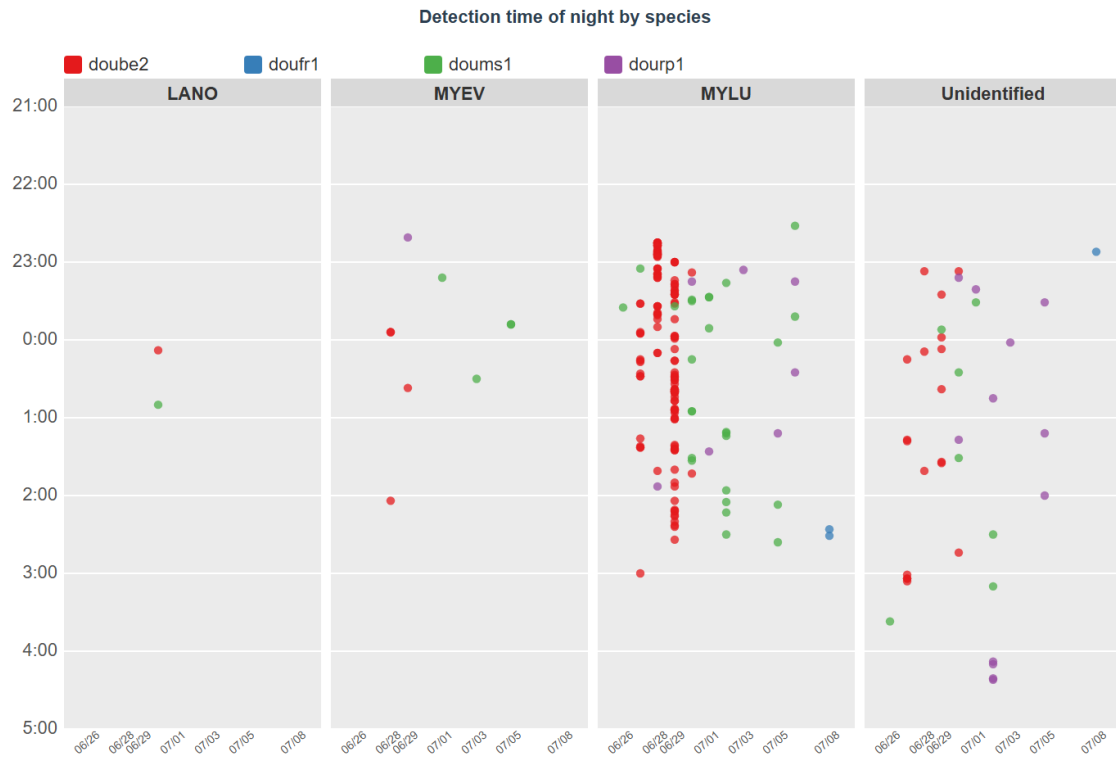


Figure 11: Nightly activity by site (2025 top, 2024 below), cell 31438 - Douglas

2025



2024

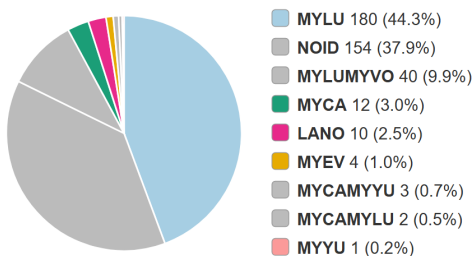


Figure 12: Detection times by species and site (2025 top, 2024 below), cell 31438 - Douglas

Grid cell 33486 - Sitka

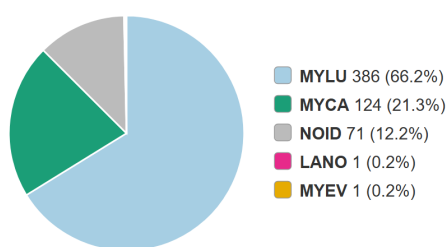
2025

stkbe3



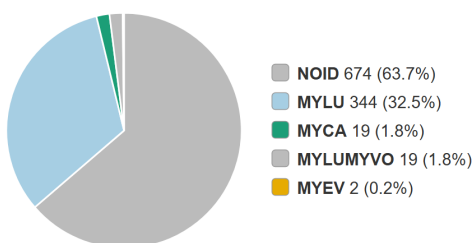
Total bats: 406

stkms2



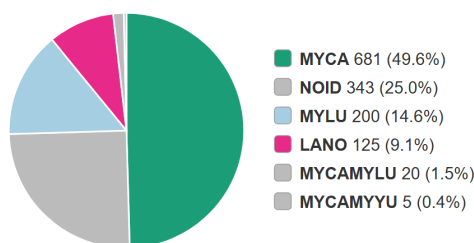
Total bats: 583

stkpdl



Total bats: 1,058

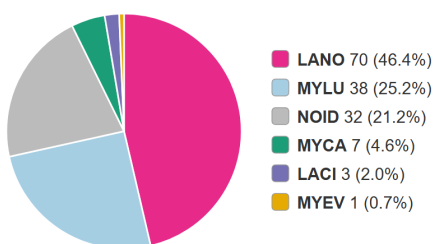
stkrp2



Total bats: 1,374

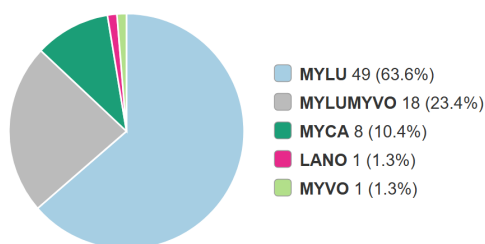
2024

stkbe3



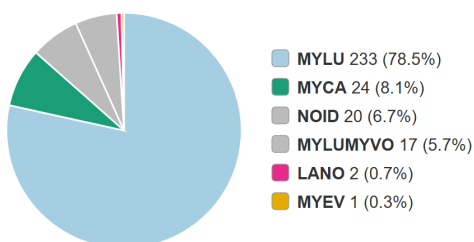
Total bats: 151

stkms2



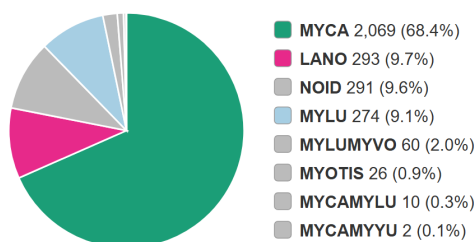
Total bats: 77

stkpdl



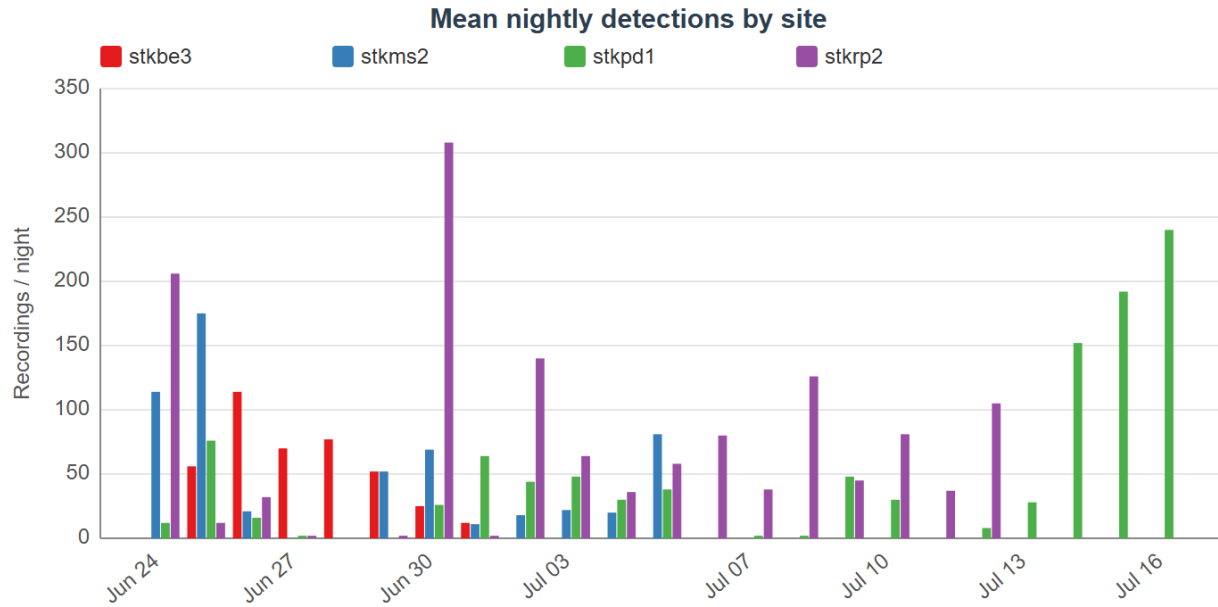
Total bats: 297

stkrp2



Total bats: 3,025

2025



2024

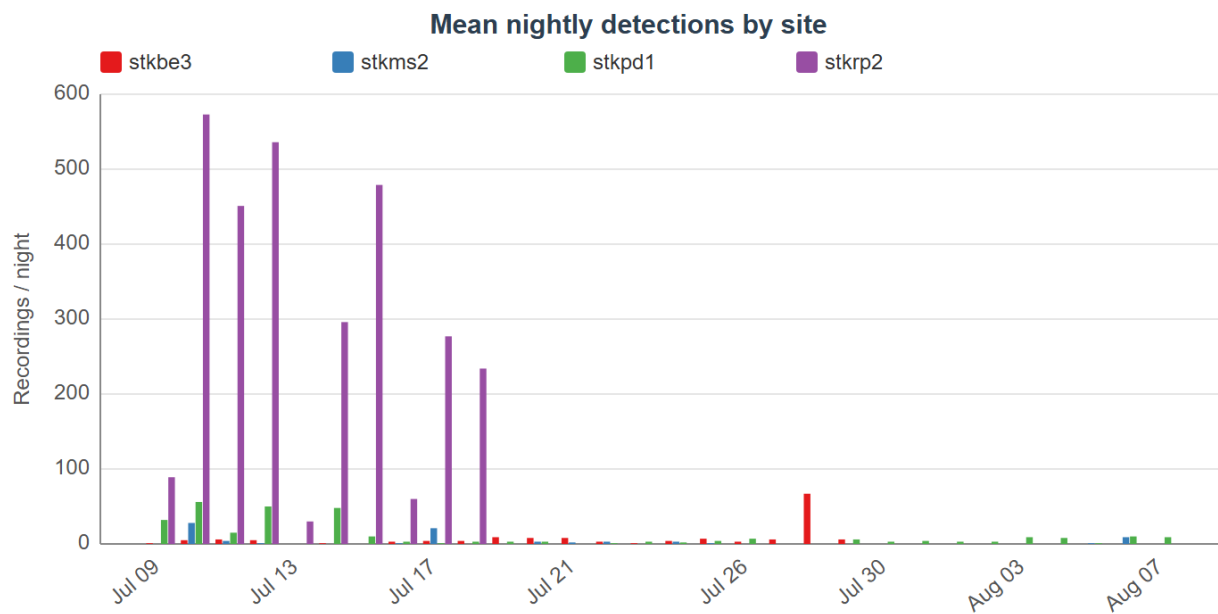
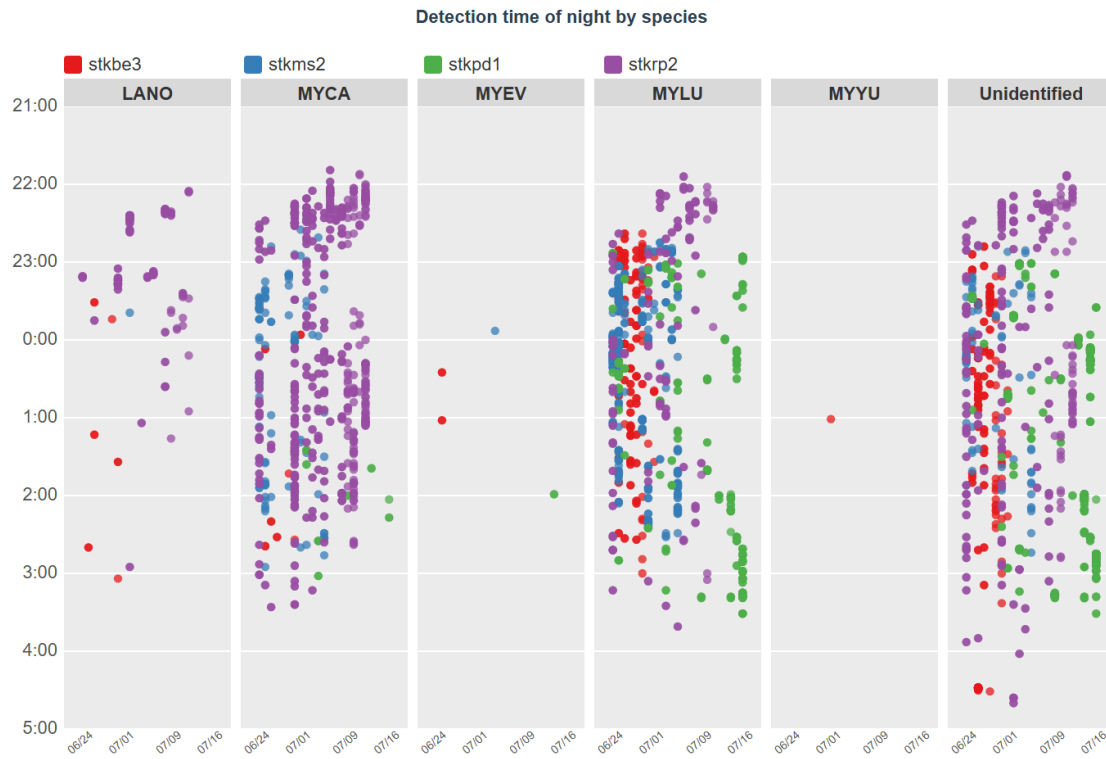


Figure 13: Nightly activity by site (2025 top, 2024 below), cell 33486 - Sitka

2025



2024

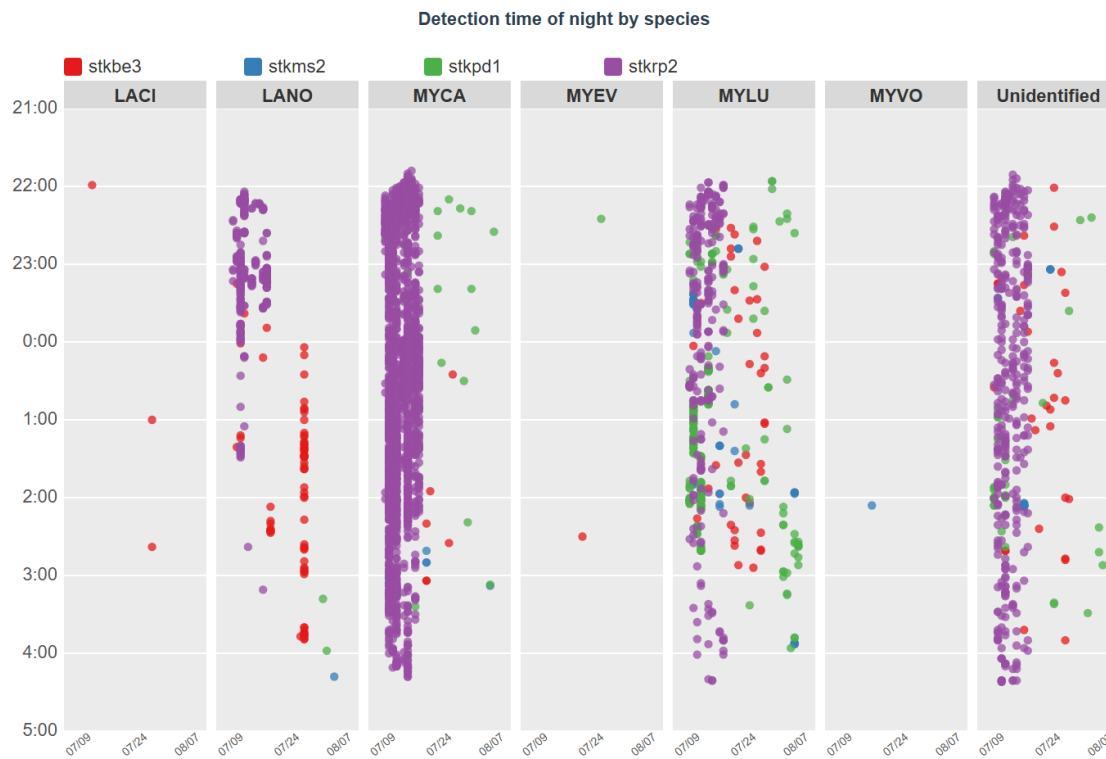
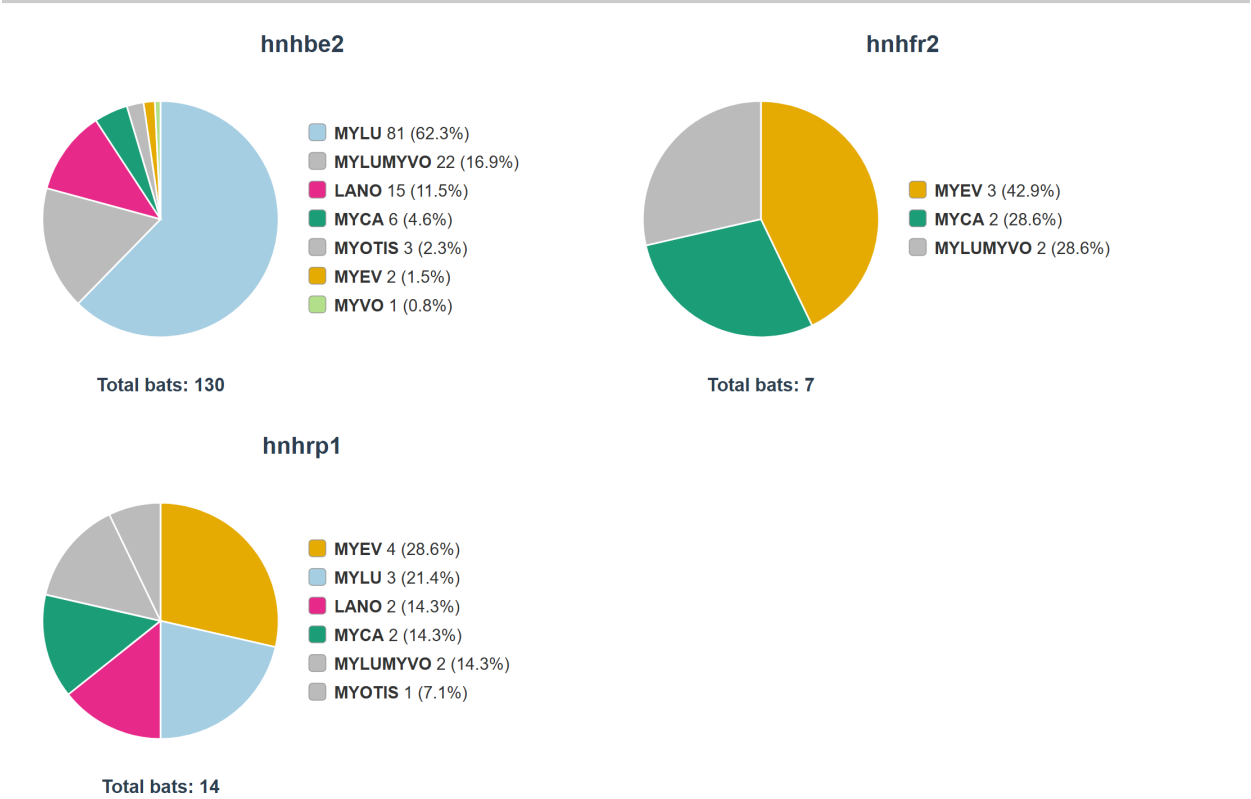


Figure 14: Detection times by species and site (2025 top, 2024 below), cell 33486 - Sitka

Grid cell 34254 - Hoonah

2024



2024

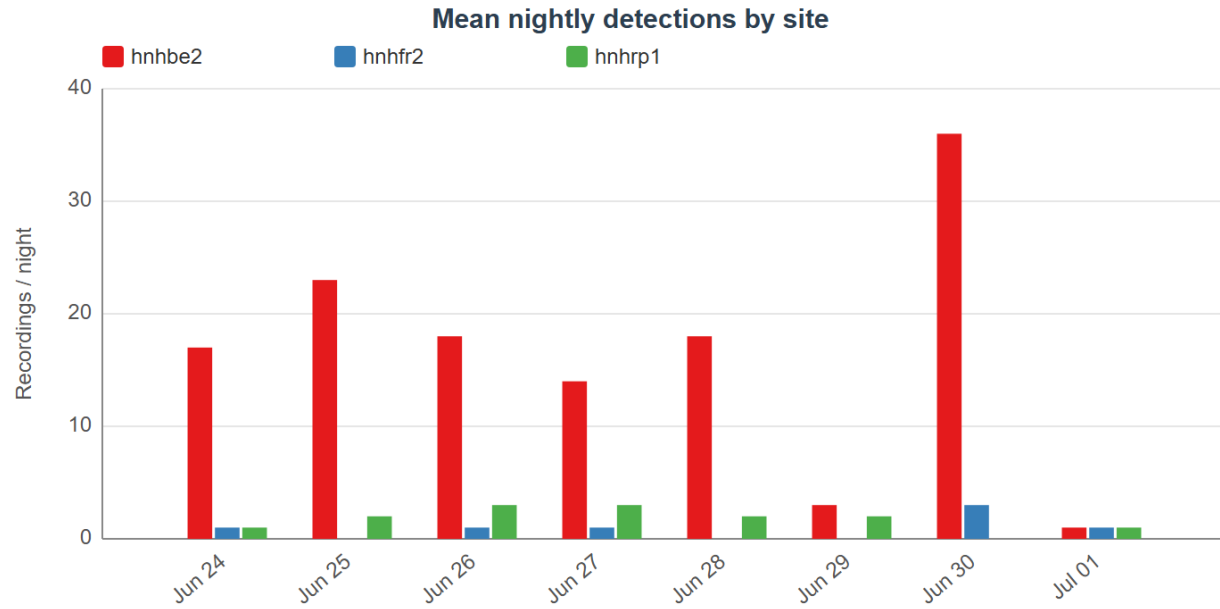


Figure 15: Nightly activity by site (2025 top, 2024 below), cell 34254 - Hoonah

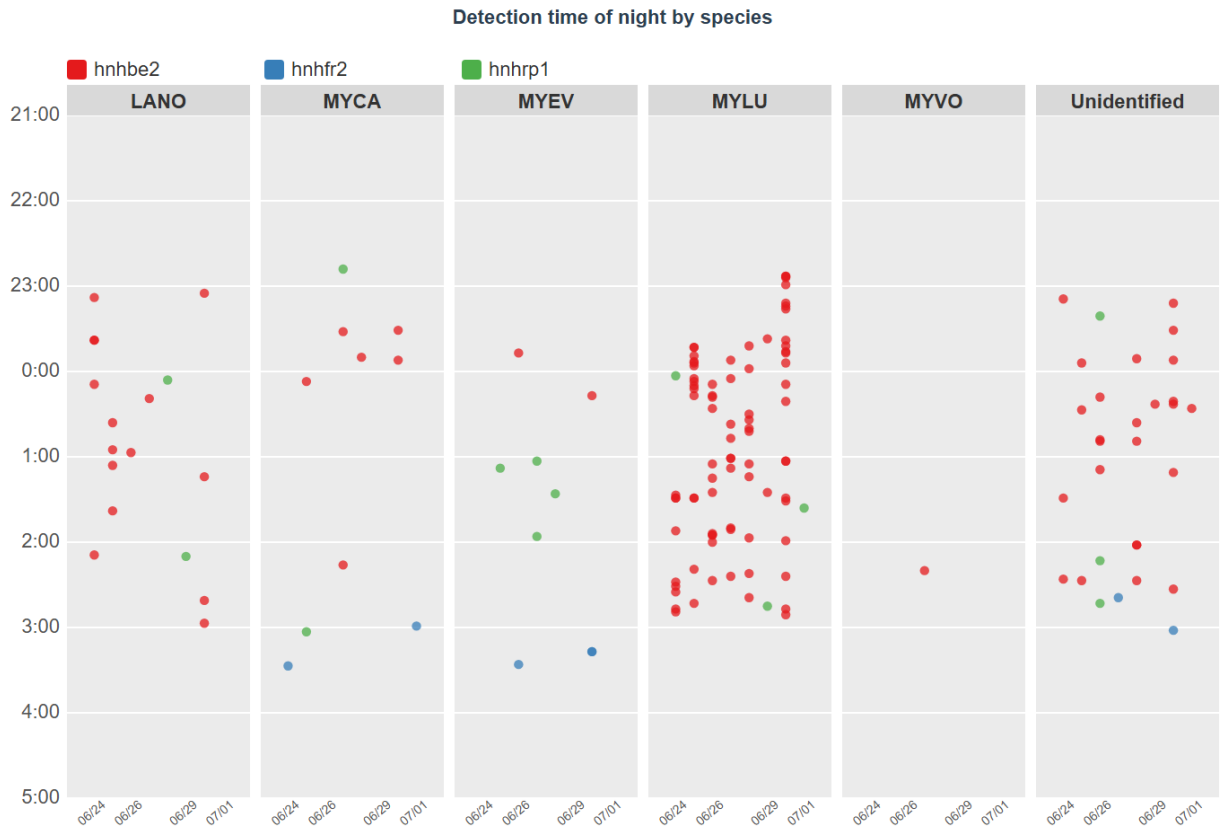
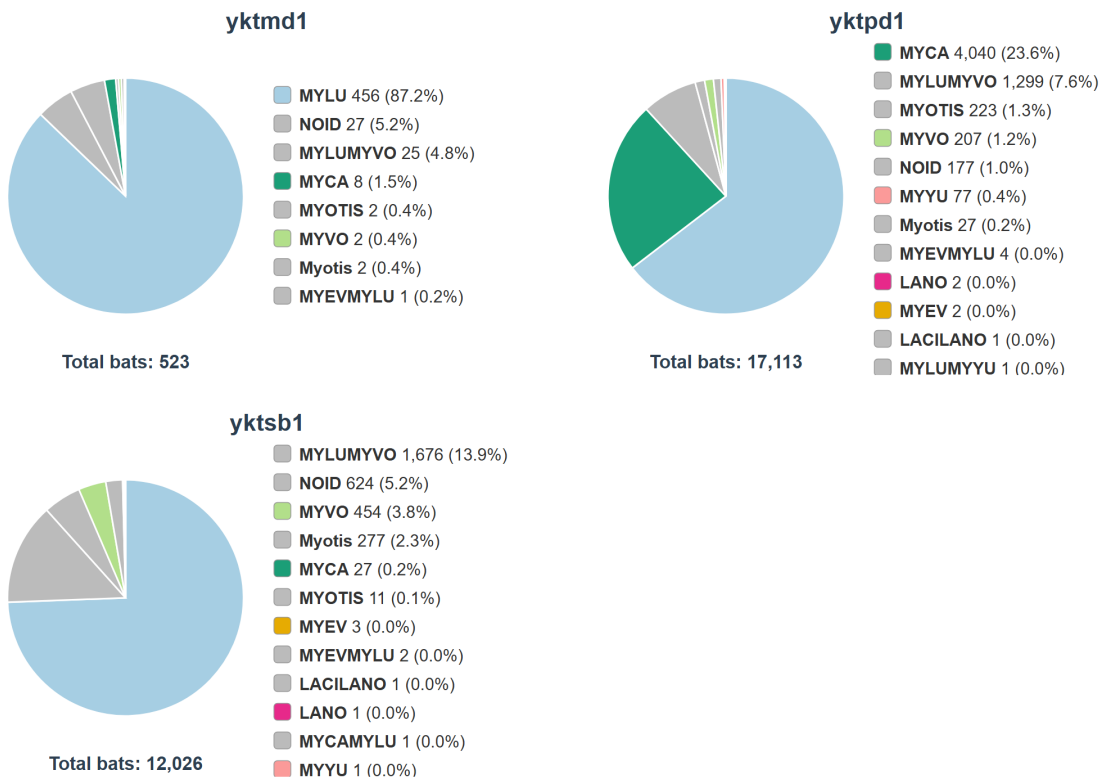


Figure 16: Detection times by species and site (2025 top, 2024 below), cell 34254 - Hoonah

Grid cell 34338 - Yakutat

2024



2024

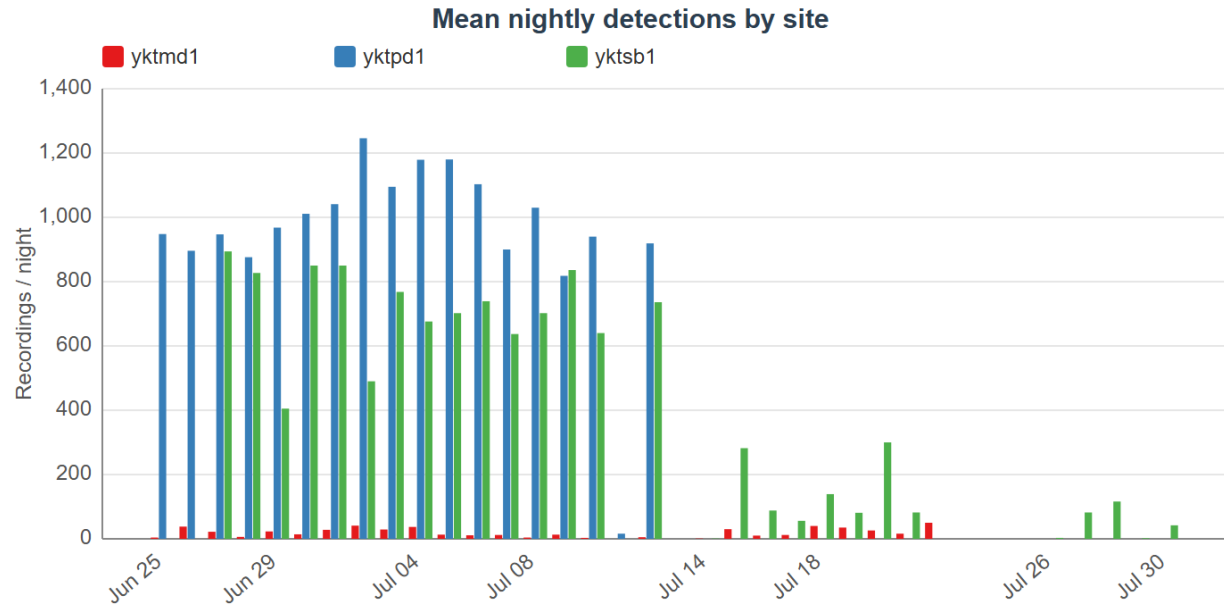


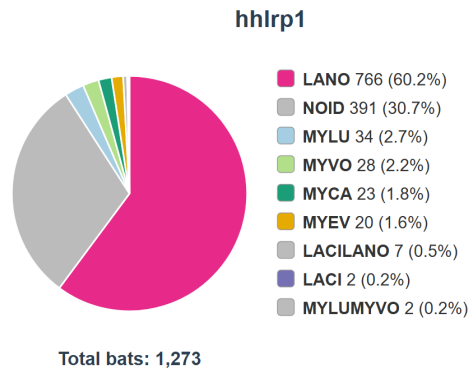
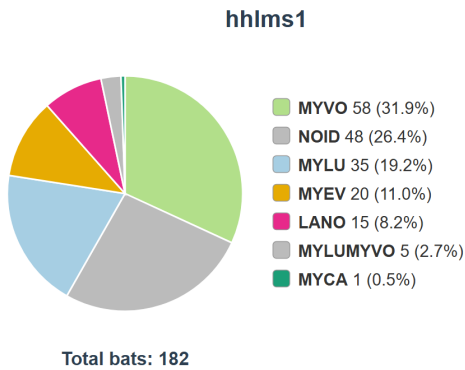
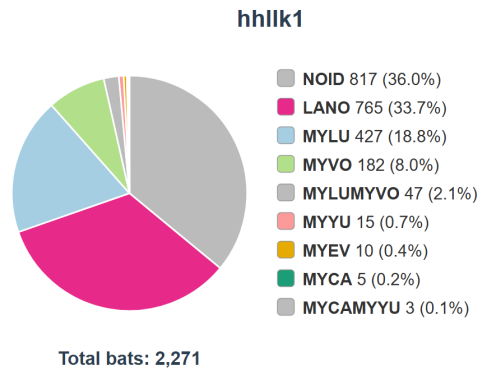
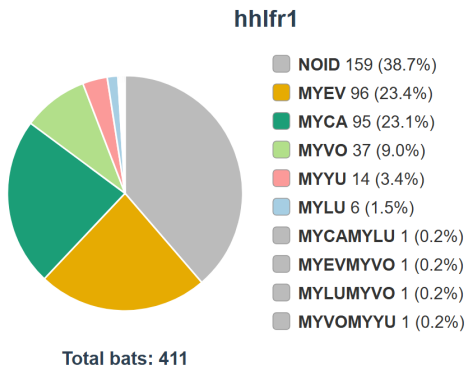
Figure 17: Nightly activity by site (2025 top, 2024 below), cell 34338 - Yakutat



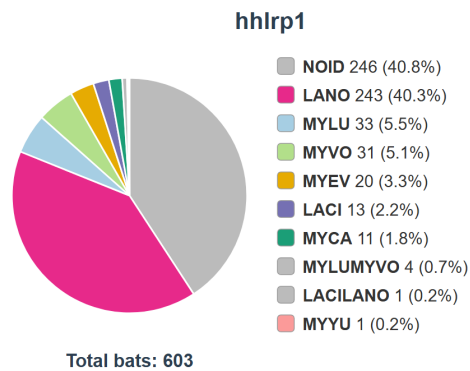
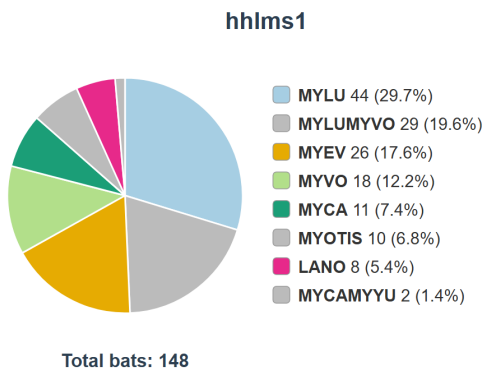
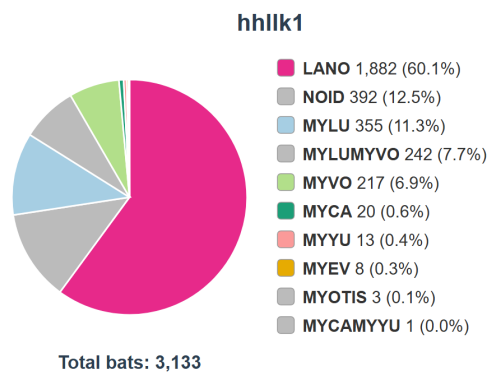
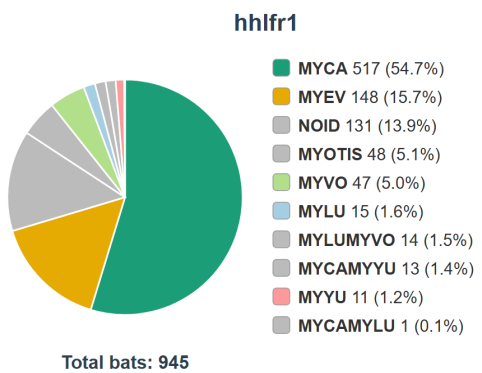
Figure 18: Detection times by species and site (2025 top, 2024 below), cell 34338 - Yakutat

Grid cell 47310 - Ketchikan

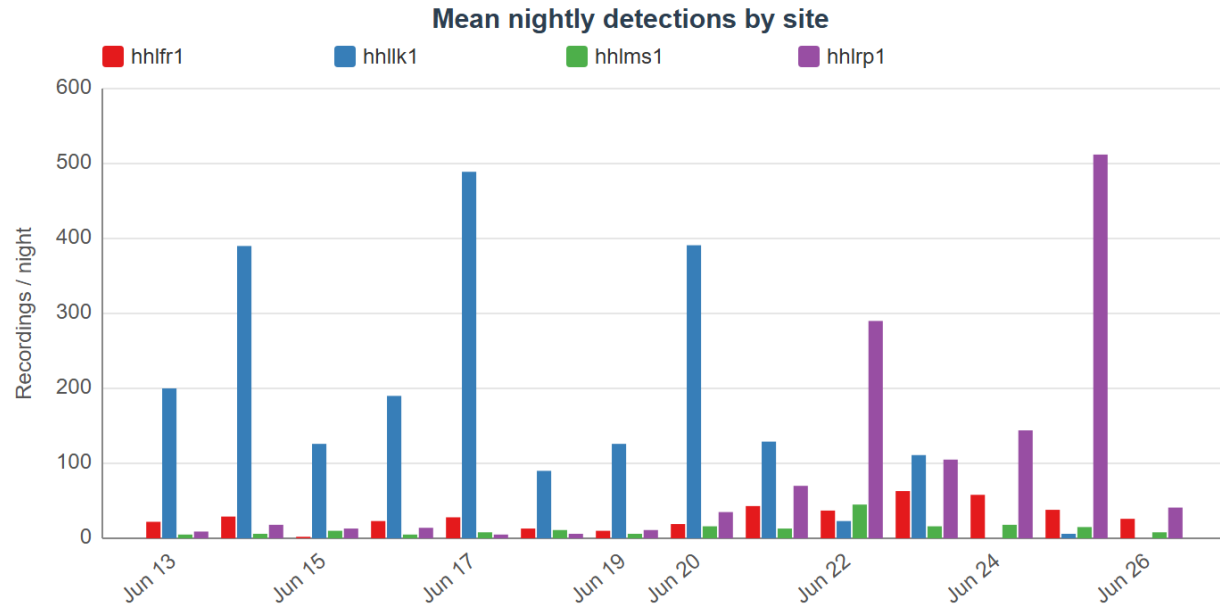
2025



2024



2025



2024

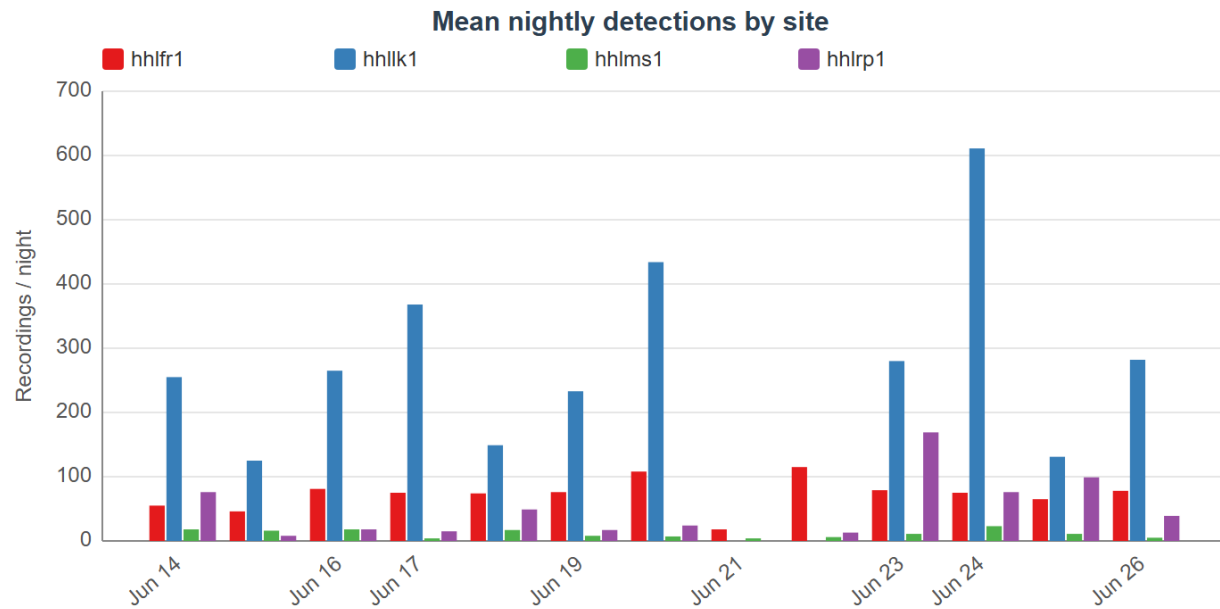
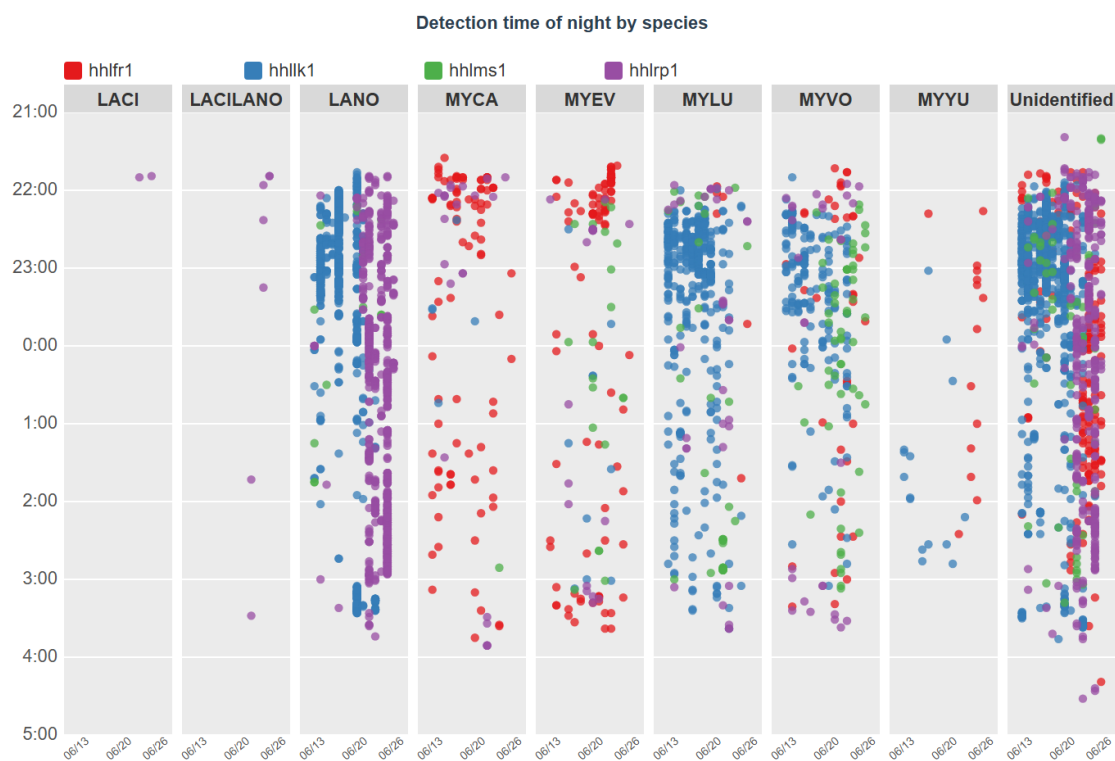


Figure 19: Nightly activity by site (2025 top, 2024 below), cell 47310 - Ketchikan

2025



2024

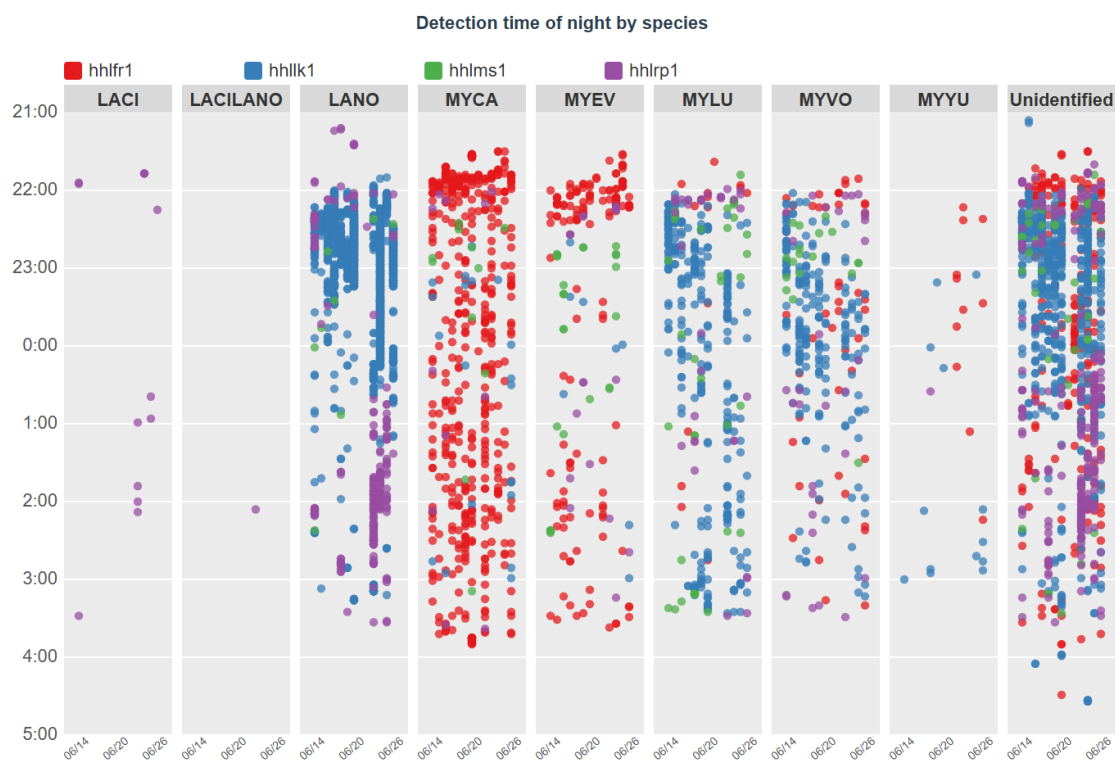
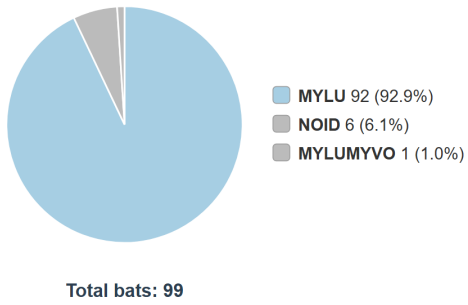


Figure 20: Detection times by species and site (2025 top, 2024 below), cell 47310 - Ketchikan

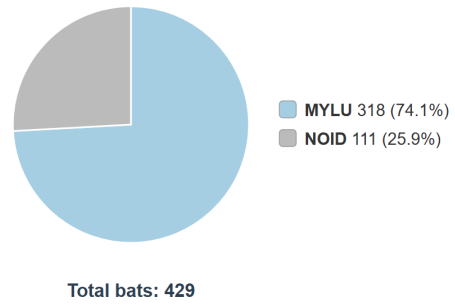
Grid cell 117173 - Anchorage

2025

Jewel Lake



Little Campbell Lake



2025

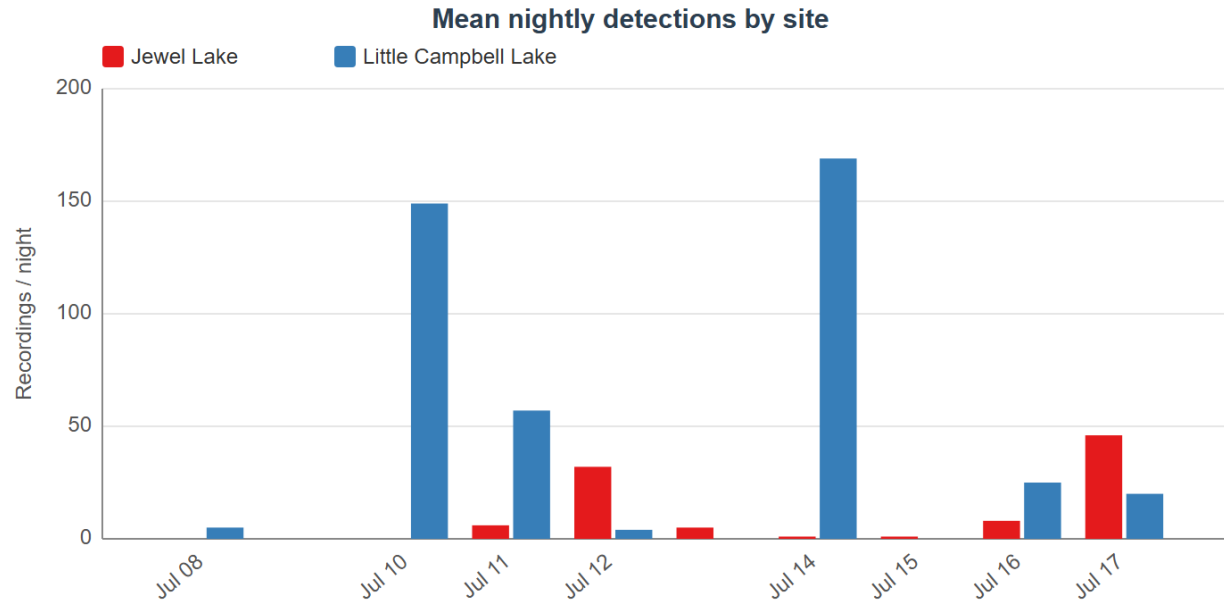


Figure 21: Nightly activity by site (2025 top, 2024 below), cell 117173 - Anchorage

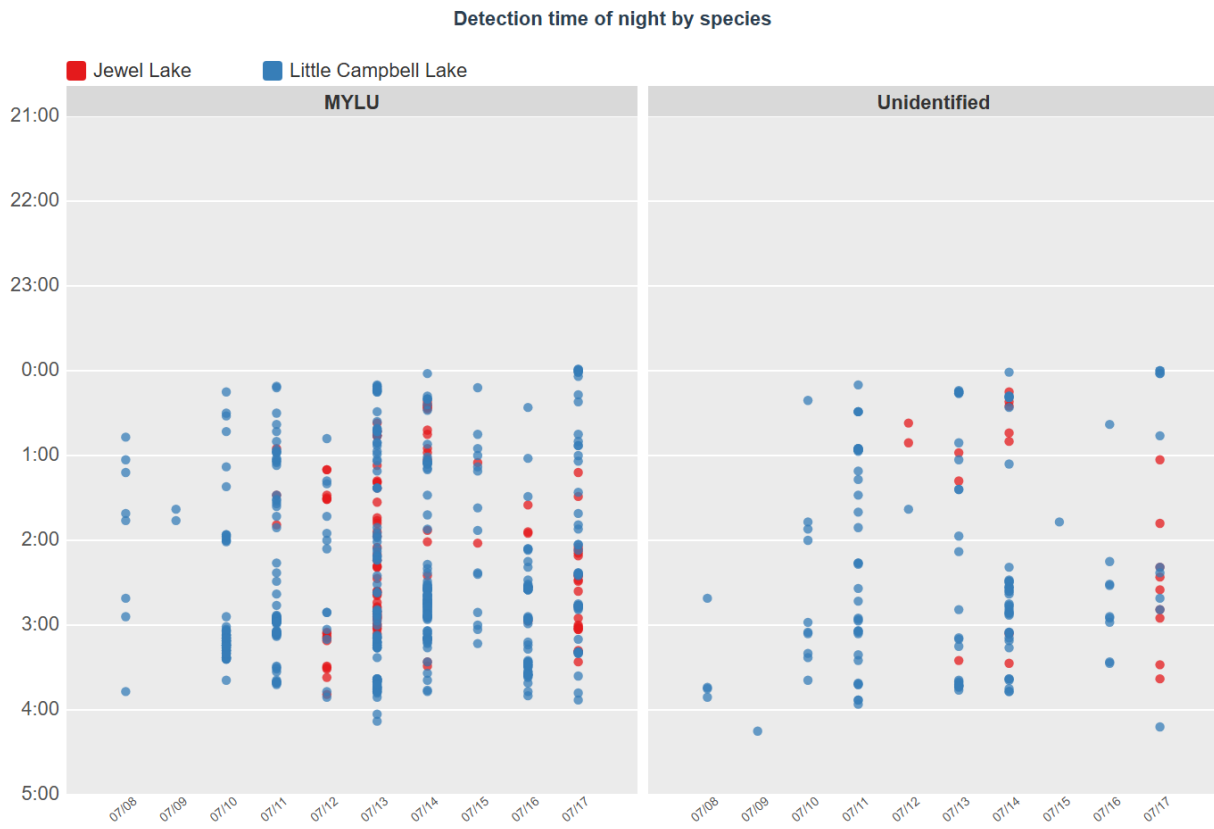
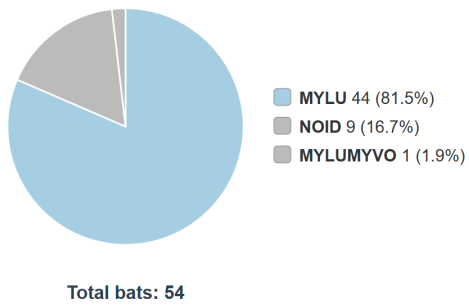


Figure 22: Detection times by species and site (2025 top, 2024 below), cell 117173 - Anchorage

Grid cell 203189 - Anchorage

2025

Basher Lake



University Lake Park

No detections

2025

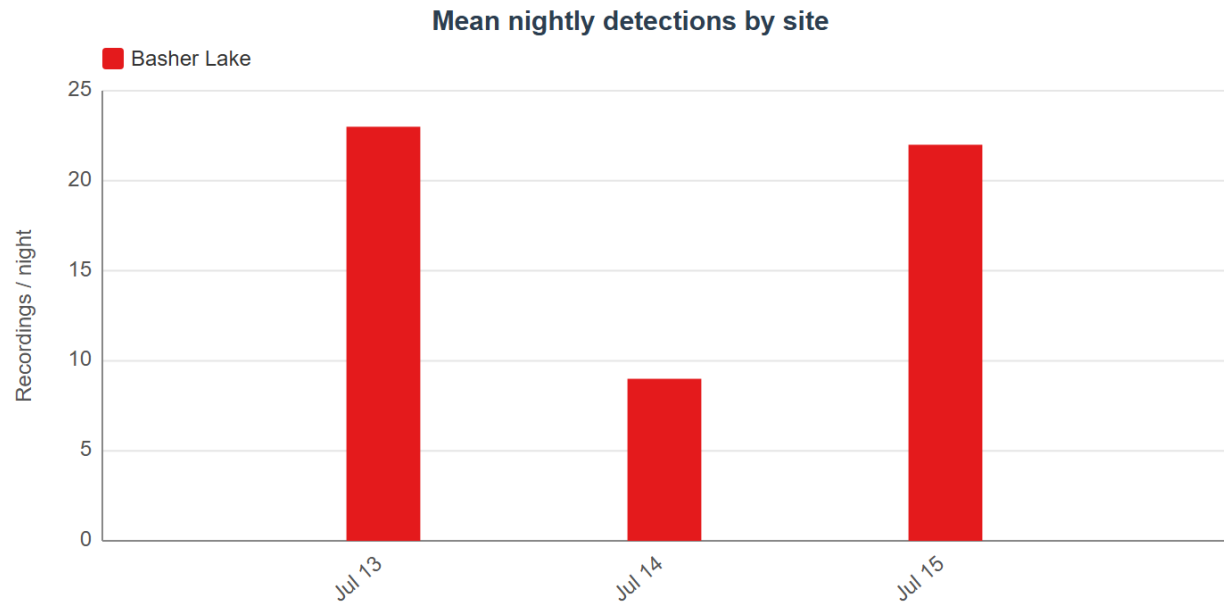
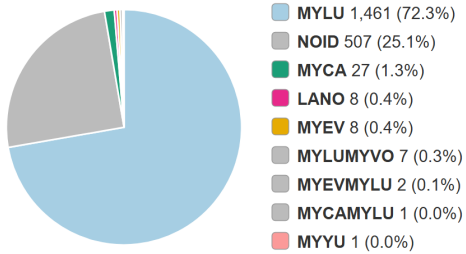


Figure 23: Nightly activity by site (2025 top, 2024 below), cell 203189 - Anchorage

Grid cell 315563 - Valley

2025

mvafmr1

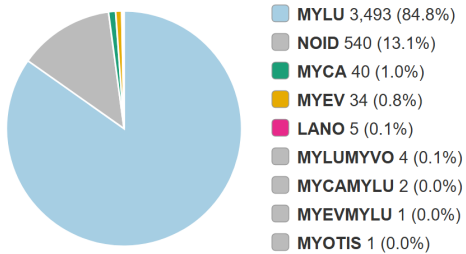


Total bats: 2,022

mvalk1

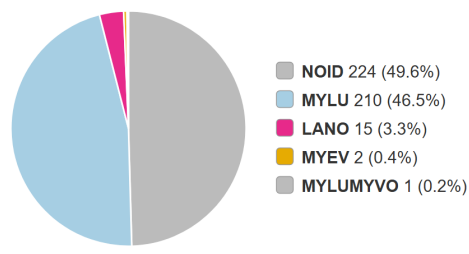
No detections

mvapd1



Total bats: 4,120

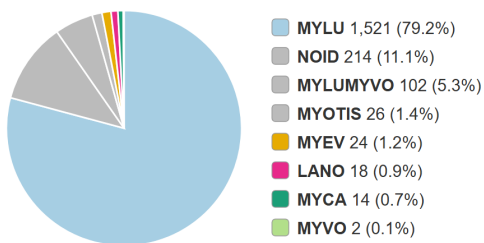
mvawt2



Total bats: 452

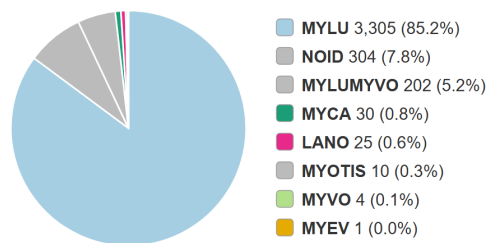
2024

mvafmr1



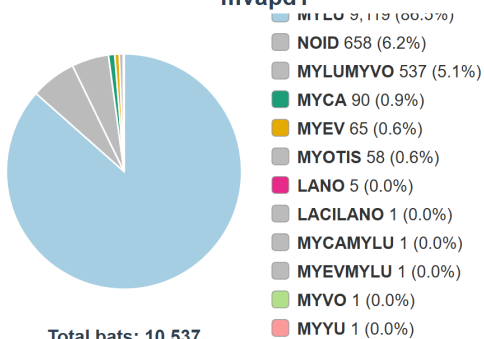
Total bats: 1,921

mvalk1



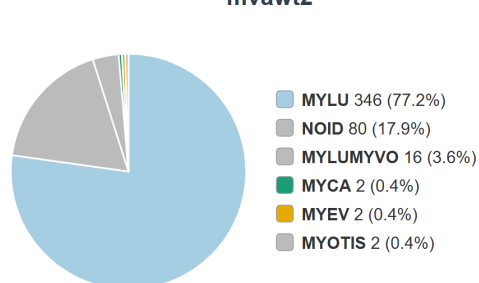
Total bats: 3,881

mvapd1



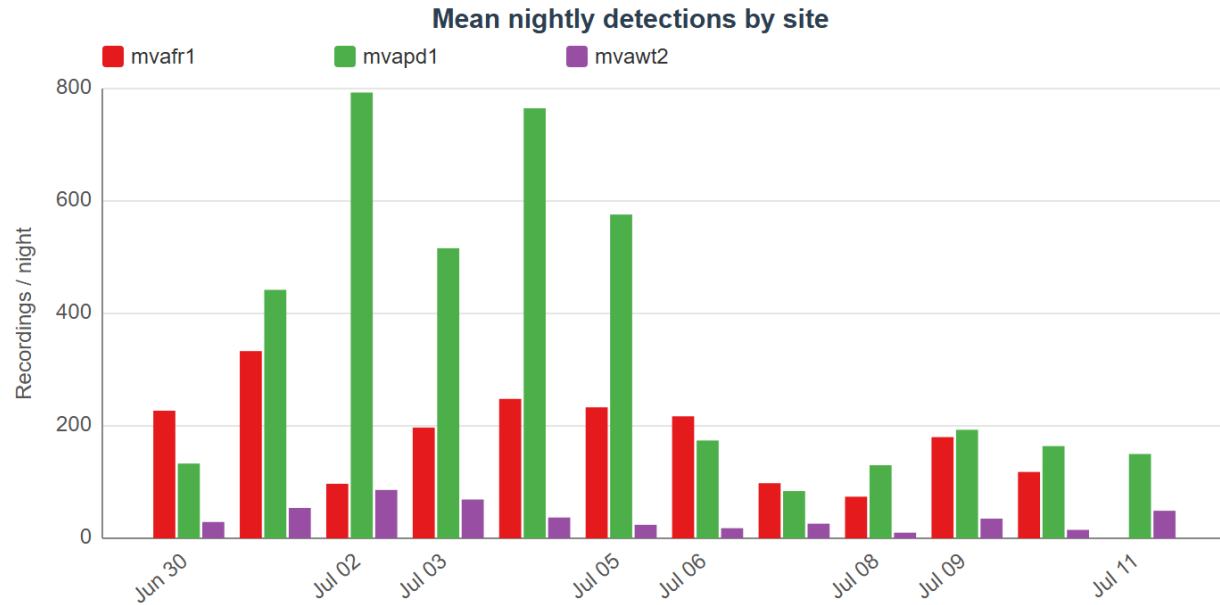
Total bats: 10,537

mvawt2



Total bats: 448

2025



2024

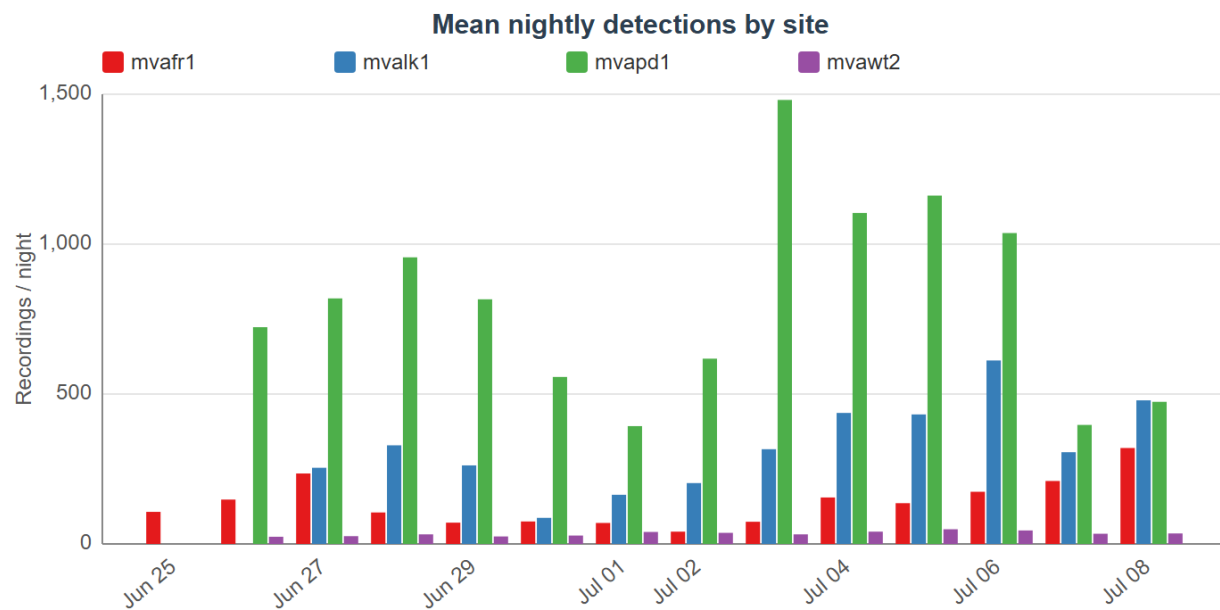
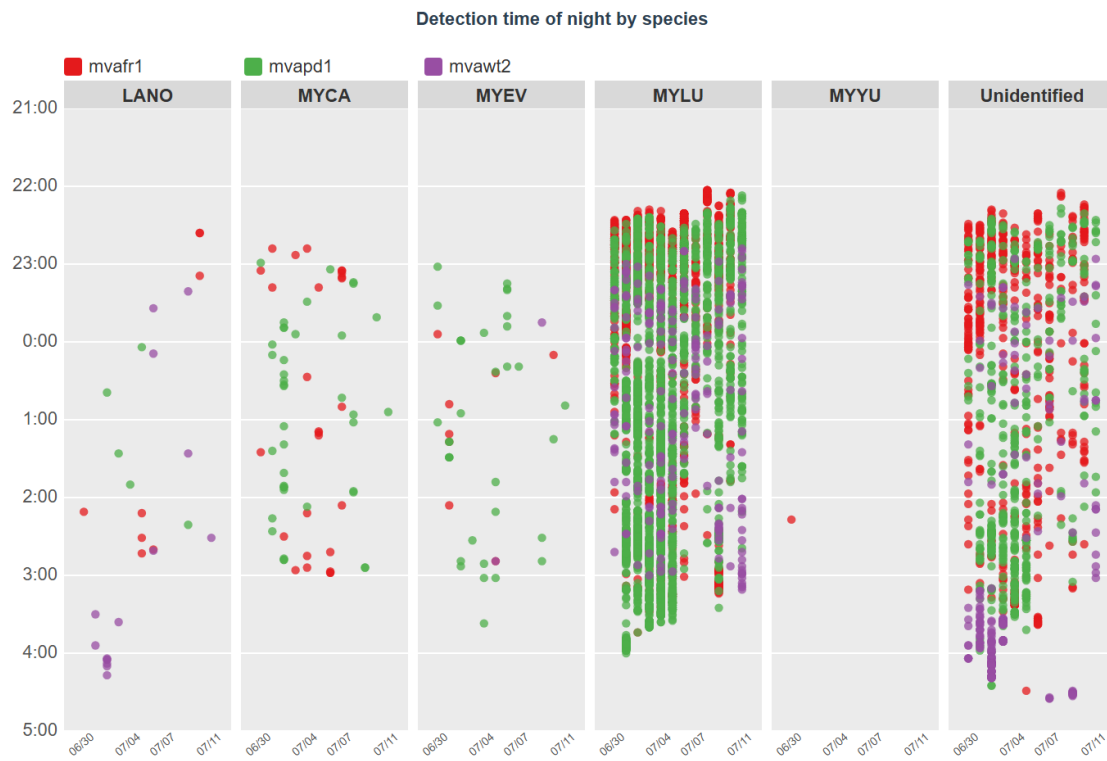


Figure 25: Nightly activity by site (2025 top, 2024 below), cell 315563 - Valley

2025



2024

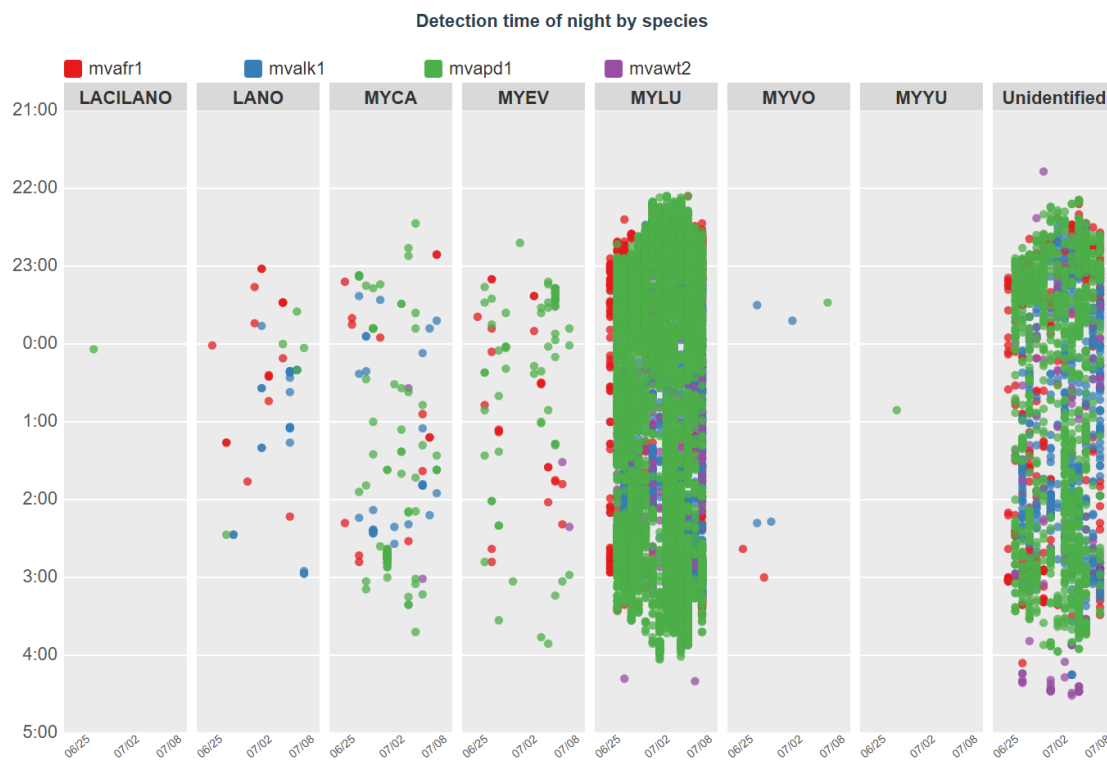


Figure 26: Detection times by species and site (2025 top, 2024 below), cell 315563 - Valley

Literature Cited

- Blejwas, Karen, Laura Beard, Joseph Buchanan, et al. 2023. “Could White-Nose Syndrome Manifest Differently in *Myotis Lucifugus* in Western Versus Eastern Regions of North America? A Review of Factors.” *Journal of Wildlife Diseases*. 59(3): 381–397 59: 381–97. <https://doi.org/10.7589/JWD-D-22-00050>.
- Blejwas, Karen, Grey Pendleton, Michael Kohan, and Laura Beard. 2021. “The Milieu Souterrain Superficiel as Hibernation Habitat for Bats: Implications for White-Nose Syndrome.” *Journal of Mammalogy* 102 (June): 1110–27. <https://doi.org/10.1093/jmammal/gyab050>.
- Cheng, Tina L., Jonathan D. Reichard, Jeremy T. H. Coleman, et al. 2021. “The Scope and Severity of White-Nose Syndrome on Hibernating Bats in North America.” *Conservation Biology* 35 (5): 1586–97. <https://doi.org/10.1111/cobi.13739>.
- Loeb, Susan C., Thomas J. Rodhouse, Laura E. Ellison, et al. 2015. *A Plan for the North American Bat Monitoring Program (NABat)*. Asheville, NC. <https://doi.org/10.2737/SRS-GTR-208>.
- MacKenzie, Darryl I., James D. Nichols, James E. Hines, Melinda G. Knutson, and Alan B. Franklin. 2003. “Estimating Site Occupancy, Colonization, and Local Extinction When a Species Is Detected Imperfectly.” *Ecology* 84 (8): 2200–2207. <https://doi.org/10.1890/02-3090>.
- Meteostat. 2026. *Meteostat: Access and Analyze Historical Weather and Climate Data with Python*. V. 2.1.4. Released. <https://pypi.org/project/meteostat/>.
- NASBR Statement on Wind Energy Impacts on Bat Populations. 2024. <https://www.nasbr.org/wind>.
- Neubaum, Daniel J. 2018. “Unsuspected Retreats: Autumn Transitional Roosts and Presumed Winter Hibernacula of Little Brown *Myotis* in Colorado.” *Journal of Mammalogy* 99 (6): 1294–306. <https://doi.org/10.1093/jmammal/gyy120>.
- Olson, Link E., and Aren M. Gunderson. 2014. “First Records of Yuma *Myotis* (*Myotis Yumanensis*) in Alaska.” *Northwestern Naturalist* 95 (3): 228–35. <https://doi.org/10.1898/13-29.1>.
- Rae, Jason, and Karen Blejwas. 2026. *North American Bat Monitoring Program in Alaska. 2015 - 2023 Data Summary and Activity Trend Analyses*. Wildlife Conservation Society Canada (WCSC).
- Rae, Jason, and Cori L Lausen. 2024. *North American Bat Monitoring Program in British Columbia*.
- Reichert, Brian, Cori Lausen, Susan Loeb, et al. 2018. *A Guide to Processing Bat Acoustic Data for the North American Bat Monitoring Program (NABat)*.
- Segers, Jordi. n.d. *CWHC-RCSF :: Canadian Wildlife Health Cooperative - Réseau Canadien Pour La Santé de La Faune*. https://www.cwhc-rcsf.ca/white_nose_syndrome_reports_and_

[maps.php#top](#).

Segers, Jordi, Scott McBurney, Megan Jones, and Damien Joly. 2025. *The Canadian Wildlife Health Cooperative Annual National Bat Health Report – 2025*.

Stratton, Christian, Kathryn M. Irvine, Katharine M. Banner, Wilson J. Wright, Cori Lausen, and Jason Rae. 2022. “Coupling Validation Effort with in Situ Bioacoustic Data Improves Estimating Relative Activity and Occupancy for Multiple Species with Cross-Species Misclassifications.” *Methods in Ecology and Evolution* 13 (6): 1288–303. <https://doi.org/10.1111/2041-210X.13831>.

Udell, Bradley J., Bethany R. Straw, Helen T. Davis, et al. 2025. *Results and Code from the North American Bat Monitoring Program’s (NABat) Integrated Species Distribution and Trend Models for Hoary Bat and Silver-Haired Bat*. <https://doi.org/10.5066/P134REO7>.

White Nose Syndrome Response Team. 2026. *White-Nose Syndrome*. <https://www.whitenosesyndrome.org/>.